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Long wave absorption by a dual purpose Helmholtz resonance OWC breakwater

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ABSTRACT

In this paper, a novel hybrid breakwater-WEC system consisting of a Helmholtz resonance oscillating water column (OWC) caisson array and a perforated wall is proposed to attenuate long waves. The principle of Helmholtz resonance OWC is analogous to the acoustic Helmholtz resonator for noise reduction. A semi-analytical model was developed based on potential flow theory using a matching eigenfunction method to investigate wave interaction with the OWC-breakwater system. The velocity singularity at the tip of the wall is resolved using the Galerkin method to improve the model convergence. The semi-analytical model of the present wave-air-structure interaction problem is verified using Haskind relation and energy conservation rule. Furthermore, we validate the model using numerical and experimental data in terms of wave reflection and wave power absorption. Hydrodynamic characteristics of the proposed hybrid breakwater-WEC system and the influence of key geometrical and wave parameters were investigated. Theoretical results indicate that long waves can be effectively absorbed by the OWC caisson through Helmholtz resonance for both normal and oblique waves. The resonance frequency of OWC varies with the chamber wall draft and chamber width in the along-shore direction. It was found that installing a perforated wall in front of the OWC array could broaden the frequency bandwidth of significant wave attenuation. The along-shore length of each OWC chamber ($2l$) and the incident wave angle (θ) are identified as the key parameters that affect the hydrodynamic performance of the system at high frequency range due to the along-shore sloshing of water column, but have a relatively minor effect on long wave attenuation. The present OWC-breakwater system demonstrates advantage and superior performance over conventional rubble mound breakwaters and breakwaters of multiple perforated walls in attenuating long waves. The findings of this paper provide guidance for a novel design of long wave absorbing breakwaters.

1. Introduction

Coastal infrastructures such as harbors and ports consume large amounts of energy to operate and are often exposed to high wave actions. In addition, there is a rising demand for more sustainable and environmentally friendly renewable energy. As a result, a novel concept of hybrid breakwater and wave energy converter (WEC) system has been proposed to save cost and space, and to serve the dual purpose of coastal protection and ocean energy extraction (Zhao et al., 2019a). Conventional breakwaters, however, are designed to work effectively for short waves mainly. Therefore, long waves may propagate behind the breakwater and severely damage the shoreline and coastal infrastructure such as port and harbor (Luijendijk et al., 2018; Bendoni et al., 2019; Leaman et al., 2021). There is a lack of proof-of-concept study on long wave attenuation capacity of breakwaters or hybrid breakwater-WEC system.

There are a large variety of breakwaters, such as vertical wall, rubble breakwater, perforated breakwater, floating breakwater, and circular breakwater. The working principle of vertical-wall breakwater is to reflect the incident waves without dissipating much wave energy, which leads to a significant reflection coefficient. The rubble breakwater, perforated breakwater, circular breakwater, and other breakwaters dissipate wave energy through wave breaking or producing turbulence with relatively small wave reflections (Muttray and Oumeraci, 2005;

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Huang et al., 2011).

Recently analytical solutions for water wave interaction with various porous structures are developed, such as multi-chamber perforated caissons (Liu et al., 2012), perforated caisson with inner sloping rubble mound (Suh et al., 2001; Liu and Faraci, 2014), slotted breakwater (Rageh and Koraim, 2010). A series of efficient analytical/numerical tools within the framework of potential flow theory have been developed to predict the hydrodynamic performance of the advanced breakwaters and have been validated against experimental data (Cho and Kim, 2007; Liu and Li, 2013; Liu and Faraci, 2014). Those analytical models are computationally efficient and physically clear. And they can be used to study the characteristics of the breakwaters, among which the interesting hydrodynamic topic includes wave reflection/-transmission/dissipation (Zhu and Zhu, 2010; Liu et al., 2016a; 2016b), wave power extraction (Lovas et al., 2010; Zhang et al., 2021; Zhao et al., 2022), influence of oblique waves (Liu et al., 2007), wave loads (Jalón et al., 2019), etc. In addition, computational fluid dynamics (CFD) method under the framework of viscous fluid theory is also widely used to examine the characteristics of breakwaters. Wave transmission, transformation, and overtopping over low-crested breakwater have been investigated using Reynolds Averaged Navier Stokes–Volume of Fluid numerical methods (Losada et al., 2005, 2008; Lara et al., 2006; Peng et al., 2009; Zou and Peng, 2011; Higuera et al., 2014). Jiang et al. (2017, 2018) conducted a series of experimental and numerical studies on wave interaction with submerged circular and quarter-circular caisson breakwaters. Explicit flow analysis can be obtained from CFD simulations, which is helpful to explore the influence of viscosity of fluids. But, due to its expensive computational cost, CFD method is rarely used to simulate the performance of breakwater array.

Despite significant attenuation of short waves towards the shore, the dissipation coefficient, i.e., the ratio of the dissipated to the incident wave energy, for long waves remains low, leading to excessive long wave reflection or transmission. In addition, long waves can be induced by the quadratic difference interaction among primary short waves. As a result, long waves may contribute more than a half of total wave energy in the nearshore region. The excessive long wave reflection back to the sea in turn may affect maritime transportation outside the port, whereas the large transmission of long waves into the harbor may cause problematic ship motion at berth (Bellotti and Franco, 2011; Maa et al., 2011; Bellotti et al., 2012; López and Iglesias, 2014; Denamiel et al., 2019) and interfere with ship and harbor operations especially for coastal areas with frequent long waves (González-Marco et al., 2008; Lee et al., 2021). Extensive studies of long waves have been motivated by the fact that long waves have significant impact on nearshore morphological change and shoreline evolution (Zou 2011; Liao et al., 2021).

In recent years, many studies have been carried out to explore the coastal protection performance of WECs due to extracting partial wave energy to reduce wave reflection/transmission effectively (Mustapa et al., 2017; Zhao et al., 2019b). Floating WECs, including flapping, oscillating buoy, and oscillating water column (OWC) types, have been proved to effectively attenuate incident waves and reduce wave reflections and transmissions (Zanuttigh and Angelelli, 2013; He and Huang, 2014; Ning et al., 2016). Various floating WEC-breakwater systems have been proposed to achieve the dual function of coastal protection and wave energy extraction (Howe et al., 2020; Saeidtehrani and Karimirad, 2021; Zhao et al., 2019b, 2021a).

Several concepts have also been proposed by integrating wave energy devices with shore-based breakwaters. It was found that an OWC-breakwater system is a promising design in terms of structural stability and economy (Naty et al., 2016). Sea trial tests have been conducted for prototypes of OWC breakwater such as Mutriku breakwater, PICO, and LIMPET (Heath et al., 2001; Torre-Enciso et al., 2009; Falcão et al., 2020). Recently, hydrodynamic characteristics (e.g., wave reflection, wave loads) of hybrid breakwater-WEC systems have been investigated extensively (Contestabile et al., 2017; Lauro et al., 2019; Pawitan et al., 2019).

It is understood that OWC device reduces the wave reflection in front of a vertical wall by converting part of the incident wave energy into pneumatic energy. He and Huang (2016) experimentally confirmed that the pile-supported OWC could effectively mitigate wave reflection from vertical walls. Xu and Huang (2018) found that, compared to a stand-alone OWC-pile device, an OWC-pile array achieved weaker wave transmission and reflection. Viviano et al.'s (2016, 2019) two-dimensional experiments indicate that the reflection coefficient of OWC breakwater is less than 0.5 for $0.06 < B/L < 0.2$, where B and L denote the width of the OWC chamber and the incident wavelength. Simonetti and Cappietti (2021) found that the function of wave absorption of OWC devices may be helpful in harbor agitation reduction.

The wave absorption characteristics of OWC device depend on the natural frequency of the water column. Zhang et al. (2021) proposed a scheme to reduce wave reflection from an OWC caisson. Their results indicate that a perforated plate in front of an OWC works effectively at a high frequency region and slightly affects the hydrodynamic efficiency of the OWC. Zhao et al. (2021b) found that, due to the constructive hydrodynamic interactions of multiple water columns, the multi-chamber device possesses the advantage of broader frequency bandwidth with qualified wave power extraction performance. Peng et al.'s (2021) experiment indicates that a hybrid breakwater-WEC system consisting of triple floating pontoons with different drafts is more effective in attenuating wave energy than that with the same drafts. Mayon et al. (2021) proposed an OWC system comprising a cylindrical OWC WEC and a parabolic wave reflector, and numerical results indicate that their design enhanced the wave power extraction performance. Gang et al. (2022) theoretically found that the dual-chamber OWC improves the wave power extraction of an OWC device embedded in coastline. Ashlin et al. (2018) experimentally investigated the performance of an array of OWC devices integrated with offshore detached breakwater. Experimental results show that wave reflection of conventional OWC breakwater becomes strong for long-period waves.

To the best of the authors' knowledge, fewer investigations were conducted on breakwaters capable of attenuating longer waves which are required to ensure safe port operations and improve the test capacity of a laboratory wave tank. Gao et al. (2021) reported a scheme of using the Bragg reflection to alleviate harbor resonance characterized by long waves. Long wave absorption may be achieved by a rubble breakwater or porous breakwater, but with a high-construction cost. The working principle of the conventional porous breakwaters is wave dissipation. An alternative breakwater with different working principle is necessary to attenuate long waves efficiently and economically. Liang et al. (2006) investigated the wave reflection of an arc-shaped wave beach, which is used to absorb waves in a deep-water wave basin. They combined a wave damping beach and a reservoir device to absorb incident wave energy. Cho and Kim (2007) found that wave absorption can be improved by adjusting the inclination of perforated plate near the free surface. Song et al. (2021) found that longer waves absorption can be achieved by placing a fixed box with gap at end of a wave flume through low-frequency gap resonance. The gap resonance frequency (i.e., location of valley reflection coefficient) is determined by the size of the box. The enhancement of long wave absorption capability requires the increment of box size (draft or breadth). In addition, since the wave dissipation depends on the triggering of the gap resonance, the frequency bandwidth with effective wave absorption may be narrower.

In field of acoustics, cavities or bottles connected to the surrounding air by a thin neck may act as a Helmholtz resonator where low-frequency sound resonance may be triggered. The resonance frequency is dependent on the geometry of the Helmholtz resonator. The dimension of the thin neck is typically much smaller than that of the cavity/bottle (Ingard, 1953). The Helmholtz resonator has been widely used in acoustic attenuation, which consists of a cavity communicating with an external duct through an orifice. It is a well-known device to reduce noise in a narrow band near its natural frequency. Since the natural frequency is determined by the size of the resonator, it is straightforward to design a Helmholtz resonator with the required natural frequency (Lal and Manchar, 1987; Cai and Mak, 2018). Considering its simple, tunable, and durable characteristics, the Helmholtz resonator has been utilized in various duct-structure systems to attenuate noise (Mak and Yang, 2000; Cai et al., 2017), such as automobile muffler structure and air-conditioning ventilation duct. In addition, the Helmholtz resonator has been applied to the field of notch filters (Akihiro et al., 2014) and ultrasonic metamaterials (Ma and Sheng, 2016).

Conventional bottom-mounted breakwaters, including rubble mound breakwaters, composite breakwaters, submerged breakwaters, and curtain-wall type breakwaters, often work in principle of reflecting or dissipating incident waves (Sawaragi, 1995). Extensive investigations have been conducted on those types of breakwaters. In this paper, alternatively, inspired by the wave absorption of OWC device and the Helmholtz resonance in acoustic attenuation, we attempt to absorb longer water waves using Helmholtz resonance of OWC at low frequency range. Compared with previous studies on OWC breakwater for short waves (Wang et al., 2022; Simonetti and Cappietti, 2021; Zhang et al., 2021), the novelty of this work includes: 1) this is the first time that an array of Helmholtz resonance of OWCs is utilized for absorbing long waves; 2) a fast converging 3-D semi-analytical model is developed by resolving the velocity singularity at the tip of the thin walls; 3) the influence of oblique waves and array effect on wave absorption capacity is investigated. Unlike Zhang et al. (2021) examined wave reflection at a conventional single-chamber OWC combined with a surface-piercing perforated barrier, here we focus on 3-D Helmholtz resonance OWC breakwater array under the action of long waves and develop an efficient hydrodynamic analysis tool and the design criteria.

The main aim of the present work is to propose and theoretically test the novel design of a Helmholtz resonance OWC breakwater to absorb longer waves for coastal protection against harsh environment. Different from a conventional OWC caisson, the gap between the chamber wall and the seabed of a Helmholtz-resonance OWC is much smaller than the water depth, i.e., satisfies $(h-d_2)^2/(Bh) \sim o(1)$, where h is water depth, d_2 is the chamber wall draft and $B = 2a$ is the chamber width (see Fig. 1 (b) and (c)), which is equivalent to the neck in a Helmholtz resonator. Thus, the working principle of the proposed Helmholtz resonance OWC is similar to that of the acoustic Helmholtz resonator. An efficient semi-analytical model is developed to solve the present wave-structure interaction problem, which is validated against published experimental and numerical data in terms of wave reflection and absorption. A comprehensive analysis of wave reflection, absorption, and dissipation is provided. Our results proved the capability of long wave absorption by Helmholtz resonance OWC. Furthermore, we combine the Helmholtz resonance OWC and conventional perforated wall to broaden the bandwidth (i.e., absorb medium and long waves). Comparisons of the wave absorption between the proposed novel structure and the classical porous breakwaters are conducted, which may be of interest to coastal engineers. To the best of our knowledge, this is the first attempt to protect the coast against medium and long waves using the principle of Helmholtz resonance OWC and perforated wall.

This paper is organized as follows: In Section 2, the mathematical model is developed, and the semi-analytical solutions are validated. In Section 3, results are presented and discussed. The hydrodynamic characteristics of OWC, the natural frequency of the water column and its contribution to wave absorption were thoroughly investigated. In Section 4, the present novel design of OWC-breakwater system is compared with the conventional breakwaters in terms of long wave attenuating capability. Finally, the conclusions are summarized in Section 5.

2. Mathematical model

2.1. Definition of the hydrodynamic problem

The graphic description of a rectangular OWC caisson array with a perforated wall is shown in Fig. 1 (a). The perforated wall is located at the weather side of the OWC chamber. Two neighboring caissons are connected by a thick bottom-mounted partition wall. A pneumatic OWC chamber is connected to the ocean waves and traps the air above, where the upward and downward motion of water column would compress and decompress the trapped air in the chamber and drive the airflow through a turbine to transfer wave energy into electric energy extracted. The symbols of each component and the fluid domain are shown in Fig. 1 (b) and (c). The origin of the Cartesian-coordinate system is located at the intersection point of an OWC chamber center and still water surface. The x-y plane is located on the still water level with the x-axis pointing oppositely towards the shoreline, the y-axis pointing along the longitudinal direction of OWC caissons, and the z-axis pointing upwards vertically.

The chamber of each OWC caisson unit consists of: a) the front wave chamber (width $2b$) formed by the perforated wall, the chamber wall of rectangular OWC caisson, and the partition wall; b) the OWC chamber (width $2a$) formed by the chamber wall, the coastal cliff, and the partition wall. The length of each OWC caisson unit is equal to $2l$.

The perforated wall of draft d_1 is placed at a distance of $2b$ from the OWC chamber wall of draft d_2. The incident wavenumber, wave amplitude, angular frequency, wavelength, wave period, water depth, incident wave angle, and gravitational acceleration are expressed as k, A, ω, L, T, h, θ, and g, respectively.

2.2. Solution derivation

Under the assumption of potential flow theory, a complex velocity potential $\Phi(x, y, z, t)$ is used to describe the wave motion. For periodic waves with an angular frequency ω, the complex velocity potential can be written as

$$\Phi(x, y, z, t) = \mathrm{Re}\left[\varphi(x, y, z)\mathrm{e}^{-\mathrm{i}\omega t}\right], \tag{1}$$

where $\varphi(x, y, z)$ is the spatial velocity potential, $\mathrm{e}^{-\mathrm{i}\omega t}$ denotes the harmonic time term, i denotes the imaginary unit, and Re[ ] denotes the real part of a complex.

The velocity potential satisfies the Laplace equation, i.e.,

$$\frac{\partial^2\varphi(x, y, z)}{\partial x^2} + \frac{\partial^2\varphi(x, y, z)}{\partial y^2} + \frac{\partial^2\varphi(x, y, z)}{\partial z^2} = 0. \tag{2}$$

and the periodic boundary condition (e.g., Linton and McIver, 2001; Porter and Evans, 1995) of

$$\varphi(x,\ y + \iota\cdot 2l,\ z) = \mathrm{e}^{2\mathrm{i}k_y\iota l}\varphi(x,\ y,\ z),\ \iota = 0, \pm 1, \pm 2, \ldots\ , \tag{3}$$

where $k_y = k\sin\theta$, and k is the positive real root of the dispersion relation of $\omega^2 = gk\tanh(kh)$. The fluid domain is divided into three subdomains Ω_j (j = 1, 2, and 3), given by Ω_1: $x \in [a + 2b, \infty)$, $z \in [-h, 0]$ and $y \in [-l, l]$; Ω_2: $x \in [a, a + 2b]$, $z \in [-h, 0]$ and $y \in [-l, l]$; Ω_3: $x \in [-a, a]$, $z \in [-h, 0]$ and $y \in [-l, l]$. The linear velocity potential φ_j in Ω_j can be divided into three parts, i.e.,

$$\varphi_j = \varphi_{\mathrm{I}} + \varphi_{\mathrm{D}}^{(j)} + p\varphi_{\mathrm{R}}^{(j)},\ \ j = 1, 2, \text{and}\ \ 3, \tag{4}$$

where $\varphi_{\mathrm{D}}^{(j)}$ denotes the diffraction velocity potential in Ω_j, $\varphi_{\mathrm{R}}^{(j)}$ denotes the radiation velocity potential in Ω_j forced by the unit pressure in the pneumatic chamber, and p denotes the air pressure implemented on the surface of OWC. φ_{I} is the incident velocity potential, which is defined as

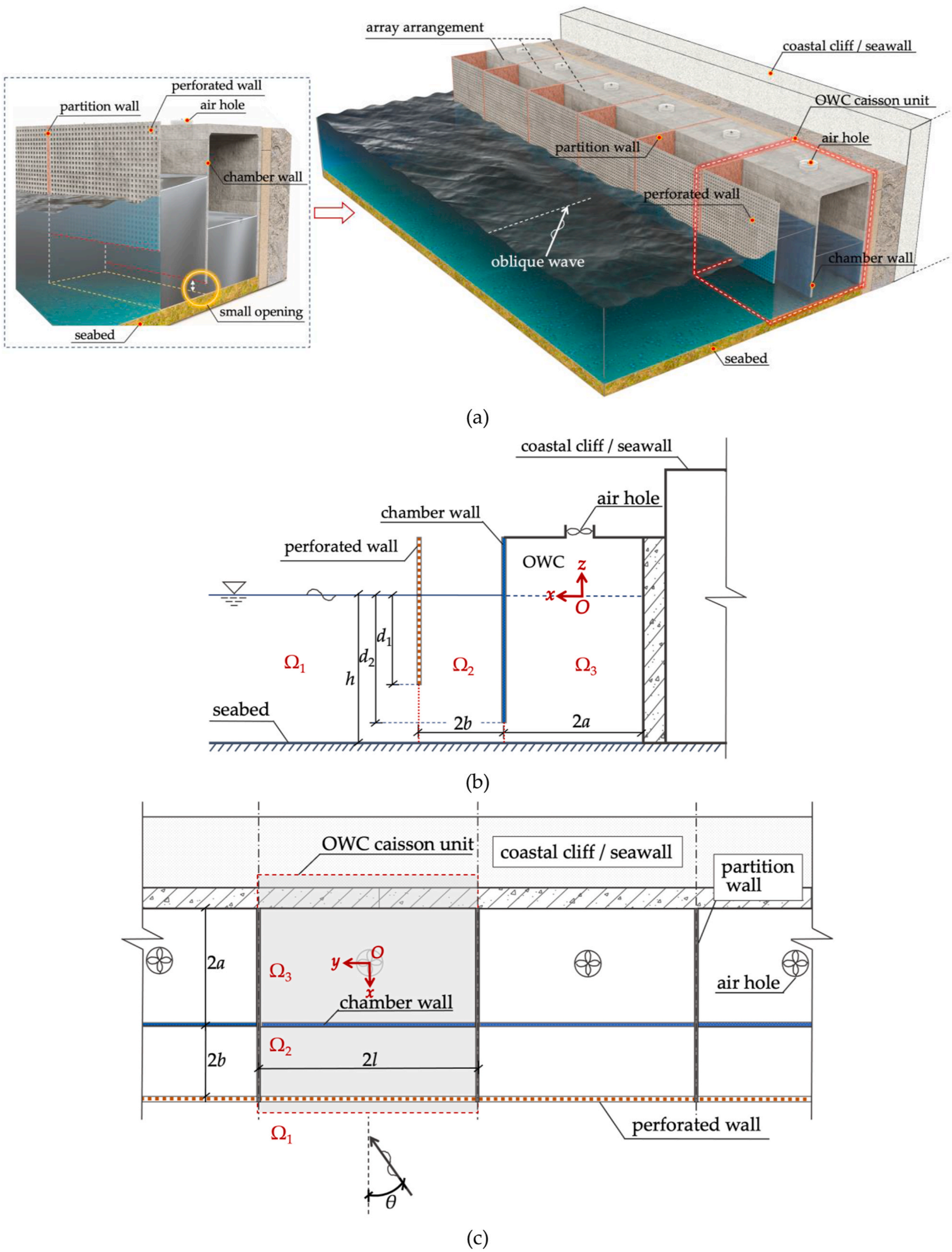


Fig. 1. Graphic description (a), side view(b), and top view (c) of the rectangular OWC caisson array with a perforated wall. Incident wave angle, the draft of perforated wall, the draft of the chamber wall, the length of an OWC unit along shoreline, width of an OWC caisson unit along x-axis direction, and width between chamber wall and perforated wall are expressed as θ, d_1, d_2, $2l$, $2a$, and $2b$, respectively.

$$\varphi_{\mathrm{I}} = -\frac{\mathrm{i}gA}{\omega}\mathrm{e}^{-\mathrm{i}k_x(x-a-2b)}\mathrm{e}^{\mathrm{i}k_y y}\frac{\cosh[k(z+h)]}{\cosh(kh)}, \tag{5}$$

where $k_x = k\cos\theta$. The linear free surface boundary condition satisfied by $\varphi_\chi^{(j)}(\chi = \mathrm{D}$ and R) is given by

$$\frac{\partial\varphi_\chi^{(j)}}{\partial z} - \frac{\omega^2}{g}\varphi_\chi^{(j)} = 0,\ x \in \Big[a, \infty),\ z = 0,\ y \in [-l, l],\ j = 1 \text{ and } 2, \tag{6}$$

and

$$\frac{\partial\varphi_{\mathrm{D}}^{(3)}}{\partial z} = \frac{\omega^2}{g}\varphi_{\mathrm{D}}^{(3)} \text{ and } \frac{\partial\varphi_{\mathrm{R}}^{(3)}}{\partial z} = \frac{\omega^2}{g}\varphi_{\mathrm{R}}^{(3)} + \frac{\mathrm{i}\omega}{\rho g},\ x \in [-a, a],\ z = 0,\ y \in [-l, l]. \tag{7}$$

The corresponding impermeable boundary condition of $\varphi_\chi^{(j)}(\chi = \mathrm{D}$ and R) at the rigid wall can be written as.

$$\begin{cases} \dfrac{\partial\varphi_\chi^{(j)}}{\partial x} = 0,\ x = a, & z \in [-d_2, 0],\ y \in [-l, l],\ j = 2 \text{ and } 3, \\ \dfrac{\partial\varphi_\chi^{(j)}}{\partial x} = 0,\ x = -a, & z \in [-h, 0],\quad y \in [-l, l],\ j = 3, \end{cases} \tag{8}$$

$$\frac{\partial\varphi_\chi^{(j)}}{\partial y} = 0,\ x \in [-a, a+2b],\ z \in [-h, 0],\ \ y = -l \text{ and } l,\ j = 2 \text{ and } 3, \tag{9}$$

and

$$\frac{\partial\varphi_\chi^{(j)}}{\partial z} = 0,\ x \in [-a, \infty),\ z = -h,\ y \in [-l, l],\ j = 1, 2 \text{ and } 3. \tag{10}$$

The far-field boundary condition of $\varphi_\chi^{(1)}(\chi = \mathrm{D}$ and R) is described as

$$\lim_{r\to\infty}\sqrt{r}\left(\frac{\partial}{\partial r} - \mathrm{i}k\right)\varphi_\chi^{(1)} = 0,\ \ r = \sqrt{x^2 + y^2}. \tag{11}$$

The boundary value problem can be formed by combining the governing equation and boundary conditions (on the free surface, impermeable boundary, and far-field boundary). The velocity potentials for each subdomain can be derived using the method of separation of variables and matching eigenfunction expansion. The expressions of diffraction and radiation potentials are given in the Appendix (i.e., Eq. (A1) and (A6)).

Following (Sollitt and Cross, 1972; Yu, 1995), the boundary condition on the thin perforated wall can be expressed as

$$\frac{\partial\varphi_\chi^{(1)}}{\partial x} = \mathrm{i}kG\left(\varphi_\chi^{(1)} - \varphi_\chi^{(2)}\right),\ x = a+2b,\ y \in [-l, l],\ z \in [-d_1, 0], \tag{12}$$

where the complex porous coefficient of the porous plate $G = \frac{\gamma}{k\delta}\left\{f - \mathrm{i}\left[1 + \frac{C_m(1-\gamma)}{\gamma}\right]\right\}^{-1}$; δ is the physical thickness of the porous plate; γ, f and C_m are the porosity, the linearized resistance coefficient, and the added-mass coefficient of the porous medium, respectively (Yu, 1995); G can be written as $G_{\mathrm{r}} + \mathrm{i}G_{\mathrm{i}}$, where G_{r} denotes the real part and G_{i} the imaginary part. Physically, G_{r} and G_{i} represent the drag and inertia effect of the porous plate, which lead to wave energy loss and phase change, respectively.

Mathematically, there exists strong velocity singularity at the tip of a perforated wall and chamber wall, which may lead to the slow convergence of the semi-analytical solutions. The multi-term Galerkin method proposed by Porter and Evans (1995) is adopted to handle the velocity singularity. According to Porter and Evans (1995), the velocity singularity at an impermeable and permeable thin wall is resolved in a different fashion. For an impermeable thin wall, the auxiliary velocity was defined at the region behind the wall (i.e., $y \in [-l, l]$, $z \in [-h, -d_2]$, $x = a$). Whereas for a permeable thin wall, the auxiliary velocity was defined at the region occupied by the wall (i.e., $y \in [-l, l]$, $z \in [-d_1, 0]$, $x = a + 2b$). Here, we introduce the velocity potential of U_χ ($\chi = \mathrm{D}$ and R) of

$$U_\chi = \sum_{m=1}^{\infty}\overline{Y}_m(y)\sum_{s_1=0}^{\infty}H_\chi^{(1)}u_{s_1}(z),\ x = a+2b,\ y \in [-l, l],\ z \in [-d_1, 0], \tag{13}$$

where $H_\chi^{(1)}(\chi = \mathrm{D}$ and R) denotes the unknowns at the perforated wall. Note that, in Porter and Evans (1995), the velocity potential is proposed for a two-dimensional problem. For the three-dimensional problem in this paper, a term of $\overline{Y}_m(y)$ is introduced in Eq. (13). $\overline{Y}_m(y)$ is given by Eq. (A2 b) in Appendix A. The auxiliary function of $u_{s_1}(z)$ is determined by

$$\widetilde{u}_{s_1}(z) = u_{s_1}(z) - \frac{\omega^2}{g}\int_{-d_1}^{z}u_{s_1}(z)\mathrm{d}z, \tag{14}$$

with

$$\widetilde{u}_{s_1}(z) = \frac{2(-1)^{s_1}\sqrt{d_1^2 - z^2}}{\pi(2s_1+1)d_1 h}Q_{2s_1}\left(-\frac{z}{d_1}\right),\ z \in [-d_1, 0]. \tag{15}$$

The function $Q_n(\ )$ denotes the nth order Chebyshev polynomial of the second kind, which can be expressed as

$$Q_n(x) = \frac{\sin[(n+1)\arccos x]}{\sin(\arccos x)}. \tag{16}$$

The orthogonal relation between the auxiliary function $u_{s_1}(z)$ and the vertical eigenfunction of $Z_u(z)$ is satisfied by

$$\int_{-d_1}^{0}u_{s_1}(z)Z_u(z)\mathrm{d}z = \frac{\mathrm{J}_{2s_1+1}(\kappa_u d_1)}{\kappa_u h},\ u = 1,\ 2,\ 3,\ \ldots \tag{17}$$

(Porter and Evans (1995), Eqs. 2.53–2.54), where the $\kappa_1 = -\mathrm{i}k$, $\kappa_u(u = 2, 3, \ldots)$ satisfies the equation of $-\omega^2 = g\kappa_u\tan(\kappa_u h)$; $Z_u(z)$ is given by Eq. (A4) in Appendix A.

In addition, we introduce the velocity V_χ to resolve the velocity singularity at the tip of the OWC chamber wall. V_χ denotes the water particle velocity beneath the chamber wall, i.e.,

$$V_\chi = \sum_{m=1}^{\infty}\overline{Y}_m(y)\sum_{s_2=0}^{\infty}H_\chi^{(2)}v_{s_2}(z),\ x = a,\ y \in [-l, l],\ z \in [-h, -d_2], \tag{18}$$

where $H_\chi^{(2)}(\chi = \mathrm{D}$ and R) denotes the unknown coefficients. The auxiliary function of $v_{s_2}(z)$ is determined by

$$v_{s_2}(z) = \frac{2(-1)^{s_2}}{\pi\sqrt{(h-d_2)^2 - (z+h)^2}}T_{2s_2}\left(\frac{z+h}{h-d_2}\right), \tag{19}$$

where $T_{2s_2}(\ )$ is the Chebyshev polynomial. The orthogonal relation between the auxiliary function $v_{s_2}(z)$ and the vertical eigenfunction of $Z_u(z)$ is given by

$$\int_{-h}^{-d_2}v_{s_2}(z)Z_u(z)\mathrm{d}z = \frac{\mathrm{J}_{2s_2}[\kappa_u(h-d_2)]}{\cos(\kappa_u h)}, \tag{20}$$

where $\mathrm{J}_{2s_2}[\,]$ is the first kind Bessel function of order $2s_2$ (see Eqs. 2.44–2.45 in Porter and Evans (1995)).

The velocity singularity caused by the rigid chamber wall and the perforated wall is resolved by introducing the Galerkin technique proposed by Porter and Evans (1995). Extensively, in this paper, by introducing the function of $\overline{Y}_m(y)$ (see Eq. (A2 b) in Appendix A), we construct the 3-D auxiliary function of U_χ and V_χ (i.e., Eqs. (13) and (18)) to handle the periodic hydrodynamic problem.

The following pressure and velocity continuity conditions of the diffraction and radiation potential at the interface of the neighboring fluid domain have to be satisfied:

1) Pressure continuity conditions:

$$\varphi_{\chi}^{(1)}-\varphi_{\chi}^{(2)}=\begin{cases}U_{\chi}, & x=a+2b,\ y\in[-l,l],\ z\in[-d_1,0],\\ 0, & x=a+2b,\ y\in[-l,l],\ z\in[-h,-d_1],\end{cases} \tag{21}$$

$$\frac{\partial\varphi_{\chi}^{(1)}}{\partial x}=\mathrm{i}kG\left(\varphi_{\chi}^{(1)}-\varphi_{\chi}^{(2)}\right),\quad x=a+2b,\ y\in[-l,l],\ z\in[-d_1,0], \tag{22}$$

and

$$\varphi_{\chi}^{(2)}=\varphi_{\chi}^{(3)},\quad x=a,\ y\in[-l,l],\ z\in[-h,-d_2]. \tag{23}$$

2) Velocity continuity conditions:

$$\frac{\partial\varphi_{\chi}^{(1)}}{\partial x}=\frac{\partial\varphi_{\chi}^{(2)}}{\partial x},\quad x=a+2b,\ y\in[-l,l],\ z\in[-h,0], \tag{24}$$

$$\frac{\partial\varphi_{\chi}^{(2)}}{\partial x}=\begin{cases}0, & x=a,\ y\in[-l,l],\ z\in[-d_2,0],\\ V_{\chi}, & x=a,\ y\in[-l,l],\ z\in[-h,-d_2],\end{cases} \tag{25}$$

and

$$\frac{\partial\varphi_{\chi}^{(3)}}{\partial x}=\begin{cases}0, & x=a,\ y\in[-l,l],\ z\in[-d_2,0],\\ V_{\chi}, & x=a,\ y\in[-l,l],\ z\in[-h,-d_2].\end{cases} \tag{26}$$

Please note that the periodic boundary condition of $\varphi(x,\ y+\iota\cdot 2l,\ z)=e^{2ik_y\iota l}\varphi(x,\ y,\ z)$ in the alongshore y-direction (see Eq. (3)) mathematically demonstrates the hydrodynamic interaction among neighboring OWC chambers. In addition, the y-direction eigenfunction of $Y_m(y)=e^{i\gamma_m y}$ with$\gamma_m=k_y+\frac{2\pi m}{2l}, m=0,\pm1,\pm2,\ldots$ (see Eq. (A2 a) in Appendix A) involved in the velocity potential of $\varphi_{\chi}^{(1)}$ contribute to the hydrodynamic interaction of the OWC array and oblique wave effect. Substituting the diffraction and radiation potentials (i.e., Eq. (A1) and (A6) in Appendix A) into the velocity and pressure continuity conditions (i.e., Eqs. (21)–(26)) at the interfaces between the adjacent subdomains, a linear system of equations can be obtained. Multiplying both sides of Eq. (24) by $Y_v^*(y)$ (complex conjugate of $Y_m(y)$) integrating with respect to y from $-l$ to l, Eqs. (21)–(23), (25) and (26) by $\overline{Y}_v(y)$ integrating with respect to y from $-l$ to l; Multiplying both sides of Eqs. (21) and (24)–(26) by $Z_u(z)$ integrating with respect to z from $-h$ to 0, Eq. (22) by $u_{f_1}(z)$ integrating with respect to z from $-d_1$ to 0, Eq. (23) by $v_{f_2}(z)$ integrating with respect to z from $-h$ to $-d_2$. The $Y_v^*(y)$, $\overline{Y}_v(y)$, $Z_u(z)$, $u_{f_1}(z)$, and $v_{f_2}(z)$ are given by Eq. (A2 a), (A2 b), (A4), (14), and (19), respectively.

We truncate m and v in the y-direction eigenfunctions to M; n and u in the z-direction eigenfunctions to N; s_1 and f_1 in the auxiliary functions to S_1; s_2 and f_2 in the auxiliary functions to S_2. Inherently, for diffraction/radiation problem, the number of unknown coefficients of the velocity potentials in Eq. (A1) or Eq. (A6) is $[(2M+1)N+M(3N+S_1+S_2+2)]$. By implementing the matching conditions (see Eqs. (21)–(26)), a set of linear equations with size of $[(2M+1)N+M(3N+S_1+S_2+2)]$ can be obtained, whose solutions denote the unknowns of the diffraction or radiation potential, detailed in Eqs. (A12)– (A14) in Appendix A. Hence, the diffraction and radiation potentials (i.e., Eq. (A1) and (A6)) can be determined.

2.3. Hydrodynamic parameters

2.3.1. Hydrodynamic efficiency

Integrating the vertical velocity of the water particle along the free surface inside the OWC chamber yields the excitation volume flux Q_e, i. e.,

$$Q_e=\int_{-a}^{a}\int_{-l}^{l}\frac{\partial\varphi_{\mathrm{D}}^{(3)}}{\partial z}\Big|_{z=0}\mathrm{d}y\mathrm{d}x=\left(-\frac{\mathrm{i}gA}{\omega}\right)\cdot 2l\cdot\sum_{n=1}^{N}\frac{1}{\overline{p}_{1n}}\left(-1+e^{2a\overline{p}_{1n}}\right)\left(A_{1n}^{(4)}+A_{1n}^{(5)}\right)[-\kappa_n\tan(\kappa_n h)], \tag{27}$$

where $\overline{p}_{1n}$ is given by Eq. (A7 b) in Appendix. Correspondingly, the radiation volume flux Q_{R} can be calculated as

$$Q_{\mathrm{R}}=\int_{-a}^{a}\int_{-l}^{l}\frac{\partial\varphi_{\mathrm{R}}^{(3)}}{\partial z}\Big|_{z=0}\mathrm{d}y\mathrm{d}x=\left(-\frac{\mathrm{i}gA}{\omega}\right)\cdot 2l\cdot\sum_{n=1}^{N}\frac{1}{\overline{p}_{1n}}\left(-1+e^{2a\overline{p}_{1n}}\right)\left(\widehat{A}_{1n}^{(4)}+\widehat{A}_{1n}^{(5)}\right)[-\kappa_n\tan(\kappa_n h)]=-(c-\mathrm{i}\mu). \tag{28}$$

where c and μ denote the radiation conductance and radiation susceptance (i.e., c = –Re[Q_{R}] and μ = Im[Q_{R}], where Im[ ] represents the imaginary part of a complex). Following Sarmento and Falcão (1985), Lovas et al. (2010), the relationship between the volume flux (Q_e and Q_{R}) and the air pressure p can be defined as

$$Q_e+pQ_{\mathrm{R}}=(c_{\mathrm{PTO}}-\mathrm{i}\mu_{\mathrm{PTO}})p, \tag{29}$$

where c_{PTO} denotes the power take-off (PTO) damping, and the air compressibility coefficient μ_{PTO} can be expressed as $\mu_{\mathrm{PTO}}=\omega V/(v^2\rho_0)$, where V denotes the initial air chamber volume of the OWC, v = 340 m/s and ρ_0 = 1 kg/m^3. The coefficient of μ_{PTO} and c_{PTO} physically incorporate the influence of the air compressibility and air turbine parameters, respectively.

For the present model, the initial air chamber volume V is often used to examine the influence of air compressibility on OWC performance following by Martins-Rivas and Mei (2008). They found that the consideration of air compressibility could broaden the bandwidth for power extraction efficiency. Note that if the air compressibility is considered (i.e., $V\neq 0$), the excitation/radiation volume flux calculated using Eqs. (27) and (28) may not be the same as that through the turbine in practice (Wang et al., 2022). Physically, the linearized air compressibility coefficient μ_{PTO} reasonably represents the spring-like effect of air compressibility.

Combing Eqs. (28) and (29), the air pressure p can be calculated by

$$p=\frac{Q_e}{-\mathrm{i}(\mu_{\mathrm{PTO}}+\mu)+(c_{\mathrm{PTO}}+c)}, \tag{30}$$

The time-averaged wave power extraction over one wave period can be expressed as

$$\overline{P}=\frac{1}{2}\frac{c_{\mathrm{PTO}}|Q_e|^2}{(c_{\mathrm{PTO}}+c)^2+(\mu+\mu_{\mathrm{PTO}})^2}. \tag{31}$$

Generally, air compressibility slightly affects the OWC dynamics for small-scale test if the pressure level is relatively lower, such as the scale of 1:50 reported in Simonetti et al. (2018). But, if the pressure level is large enough, neglecting the air compressibility may lead to an overestimation of air pressure in the OWC chamber and the volume flux, and an underestimation of power extraction efficiency (Simonetti et al., 2018). For the large-scale or prototype test, air compressibility may affect the air-water interaction and the OWC dynamics significantly (Dimakopoulos et al., 2017; Viviano et al., 2019). In this paper, we use V = 0.4hla to represent the spring-like effect of air compressibility. Since the model adapted here is derived based on an isentropic pressure-density relation, the complex air-water interaction and the viscous losses may not be resolved (Sarmento and Falcão, 1985).

The optimal PTO damping (Martins-Rivas and Mei, 2008) can be yielded as

$$c_{\mathrm{opt,PTO}} = \sqrt{c^2 + (\mu + \mu_{\mathrm{PTO}})^2}. \tag{32}$$

In this paper, we use the optimal PTO damping $c_{\mathrm{opt,PTO}}$ to calculate the extracted wave power. Therefore, the extracted wave power for the optimal PTO damping becomes

$$\overline{P} = \frac{1}{4}\frac{|Q_e|^2}{c + \sqrt{c^2 + (\mu + \mu_{\mathrm{PTO}})^2}}, \tag{33}$$

and the hydrodynamic efficiency η is given by

$$\eta = \frac{2\overline{P}}{\rho g A^2 C_g}, \tag{34}$$

where C_g is the group velocity of the incident wave

$$C_g = \frac{\omega}{2k}\left[1 + \frac{2kh}{\sinh(2kh)}\right]. \tag{35}$$

The incident potential, diffraction potential, and the radiation potential caused by the air pressure contribute to the wave amplitude ζ. Based on the linear free surface condition, wave amplitude ζ_j at region Ω_j (j = 1, 2, and 3) can be expressed as

$$\frac{\zeta_j}{A} = \left|\frac{\mathrm{i}\omega}{gA}\left(\varphi_{\mathrm{I}} + \varphi_{\mathrm{D}}^{(j)} + p\varphi_{\mathrm{R}}^{(j)}\right)\Big|_{z=0}\right|, \; j = 1,\ 2 \ \text{and} \ 3, \tag{36}$$

and $\zeta = \zeta_1 \cup \zeta_2 \cup \zeta_3$, characterizing the wave amplitude distributions of the whole domain.

2.3.2. Reflection and dissipation coefficients

Considering the scattered waves and radiated waves, the incident wave energy flux and the total reflected wave energy flux can be written as

$$F^{\mathrm{I}} = \rho g A^2 l C_g \cos\theta, \tag{37}$$

and

$$F^{\mathrm{R}} = \rho g A^2 l C_g \sum_{m=-M_1}^{M_2} \left(A_{m1}^{(1)} + p\widehat{A}_{m1}^{(1)}\right)^2 \cos\theta_m, \tag{38}$$

respectively.

The reflection coefficient K_r is defined as the square root of the ratio of the total reflected wave energy fluxes to the incident wave energy flux, i.e.,

$$K_r = \sqrt{\frac{\sum_{m=-M_1}^{M_2} \left(A_{m1}^{(1)} + p\widehat{A}_{m1}^{(1)}\right)^2 \cos\theta_m}{\cos\theta}} = \sqrt{\frac{\sum_{m=-M_1}^{M_2} \left(A_{m1}^{(1)} + p\widehat{A}_{m1}^{(1)}\right)^2 \sqrt{1 - \left(\sin\theta + \frac{\pi m}{kl}\right)^2}}{\cos\theta}}. \tag{39}$$

Based on the energy conservation law, the dissipation coefficient can be calculated as

$$K_d = 1 - \eta - K_r^2. \tag{40}$$

2.4. Model verification

2.4.1. Convergence test

To examine the convergence of the semi-analytical solution, the reflection coefficients K_r at different truncated numbers of M, N, S_1 and S_2 are calculated and listed in Table 1. The frequency range of $kh = 0.5$–4 with an interval of 0.5 is considered. The geometrical and wave parameters are fixed as $d_1/h = d_2/h = 0.9$, $a/h = b/h = 1/4$, $l/h = 0.5$, $G = 0.5\cdot(1+1\mathrm{i})$, and $\theta = \pi/6$. Table 1 indicates that, for $kh = 0.5$–4, the semi-analytical solution converges with three decimal accuracy at $M = 5$, $N = 40$, $S_1 = 5$, and $S_2 = 5$, which are used in the following calculations.

2.4.2. Haskind relationship

Based on the Haskind relationship, the excitation volume flux Q_e can also be obtained from the radiation potential and incident wave potential at far field, which is referred to as the indirect method hereinafter (Zhao et al., 2022). The excitation volume flux $Q_e^{(\mathrm{R})}$ calculated using the indirect method is given by

$$Q_e^{(\mathrm{R})} = 4\rho g l A^2 C_g \widehat{A}_{01}^{(1)} \cos\theta. \tag{41}$$

Fig. 2 shows the comparisons between Q_e calculated from Eq. (27) and $Q_e^{(\mathrm{R})}$. The solutions of the scattering and radiation velocity potentials

Table 1
Results of reflection coefficient K_r with the different truncated numbers M, N, S_1, and S_2.

Truncated numbers		Reflection coefficient K_r							
		$kh = 0.5$	$kh = 1$	$kh = 1.5$	$kh = 2$	$kh = 2.5$	$kh = 3$	$kh = 3.5$	$kh = 4$
M ($N = 40$,	2	0.490	0.417	0.318	0.297	0.368	0.474	0.346	0.505
$S_1 = 5$,	3	0.490	0.417	0.317	0.297	0.369	0.475	0.355	0.489
$S_2 = 5$)	4	0.490	0.417	0.317	0.297	0.370	0.476	0.357	0.487
	5	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	10	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	15	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
N ($M = 5$,	10	0.552	0.099	0.240	0.268	0.365	0.476	0.376	0.484
$S_1 = 5$,	20	0.520	0.333	0.296	0.292	0.369	0.476	0.361	0.486
$S_2 = 5$)	30	0.500	0.396	0.311	0.296	0.370	0.476	0.359	0.486
	40	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	50	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	60	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
S_1 ($M = 5$,	2	0.491	0.417	0.318	0.297	0.370	0.477	0.359	0.490
$N = 40$,	3	0.491	0.417	0.317	0.297	0.370	0.476	0.358	0.486
$S_2 = 5$)	4	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	5	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	7	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	10	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
S_2 ($M = 5$,	2	0.489	0.417	0.316	0.297	0.370	0.476	0.358	0.486
$N = 40$,	3	0.489	0.417	0.316	0.297	0.370	0.476	0.358	0.486
$S_1 = 5$)	4	0.489	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	5	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	7	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486
	10	0.490	0.417	0.317	0.297	0.370	0.476	0.358	0.486

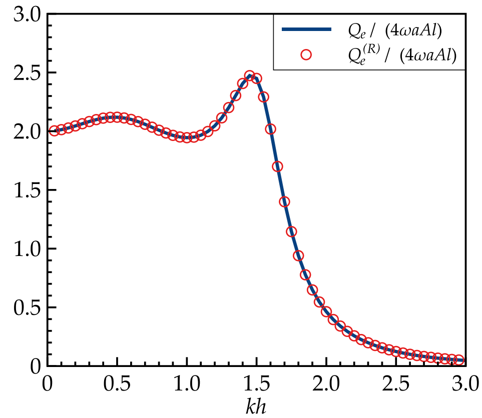


Fig. 2. Comparisons of Q_e and $Q_e^{(R)}$ against kh. The fixed parameters are: $d_1/h = 0.7$, $d_2/h = 0.5$, $a/h = 0.25$, $b/h = 0.25$, $l/h = 0.5$, $G = 0.5(1+1i)$, $\theta = \pi/4$, and $h = 10$m.

are validated by the good agreement between Q_e and $Q_e^{(R)}$ shown in Fig. 2.

2.4.3. Energy conservation law

According to the potential flow theory, the relationship of $\eta + K_r^2 = 1$ must be satisfied if there is no energy dissipation. Based on wave energy flux conservation, we verify the present model using the formula of $\eta + K_r^2 = 1$ for cases of $G = \varepsilon$ (1+1i), ε = 1E-7 and 1E7. The geometrical and wave parameters are fixed as $h = 10$m, $d_1/h = d_2/h = 0.9$, $a/h = b/h = 1/4$, $l/h = 0.5$ and $\theta = \pi/6$. As indicated in Table 2, the wave energy flux conservation is well satisfied, which also verifies derivation of the reflection coefficient and the hydrodynamic efficiency.

2.4.4. Validation against CFD results: reflection coefficient of OWC caisson

Here, we validate the present model by comparing it with viscous-flow-based numerical data (i.e., reflection coefficient) of the OWC caisson (Simonetti and Cappietti, 2021). The geometrical and wave parameters are: OWC chamber wall draft $d_2/h = 0.5$, width of the OWC chamber $2a/h = 1$, and incident wave angle $\theta = 0$. The sketch for the OWC caisson is shown in Fig. 3 (b). From the comparisons shown in Fig. 3 (a), we found that the two aspects show good consistency in trends.

2.4.5. Validation against experimental and numerical results: hydrodynamic efficiency of OWC caisson

Ning et al. (2015, 2016) presented the experimental and potential-flow-based numerical results (with and without viscous correction) of the hydrodynamic efficiency of an OWC caisson. Comparisons of the hydrodynamic efficiency η against dimensionless wavenumber kh between the present results and the experimental/numerical results are shown in Fig. 4 (a). The geometrical and wave parameters are: $h = 0.8$m, $d_2/h = 0.175$, $2a/h = 0.6875$, and $\theta = 0$. The sketch for key geometrical parameters is given in Fig. 4 (b). The good agreement shown in Fig. 4 (a) gives the confidence that the present semi-analytical model can efficiently predict the trend of the hydrodynamic efficiency of an OWC caisson.

2.4.6. Validation against experimental and CFD results: reflection coefficient of perforated/slotted wall

Here we validated our model by comparing with a perforated/slotted wall breakwater located at the weather side of the seawall. Kondo (1979) and Zhu and Chwang (2001) measured the reflection coefficient of the perforated wall with circular orifices and slotted wall, respectively. Comparisons of the calculated results and the experimental data are shown in Fig. 5 (a) and (c). The selection of physical parameters of resistance coefficient f and added-mass coefficient C_m is in accordance with Li et al., (2006)'s investigation. It can be concluded that calculations and the experimental data are generally in good agreement, even though there exists a slight discrepancy between the experimental data and the present results. The discrepancy is due to the viscosity neglected in the present potential-flow solver. In conclusion, the agreement of the trend of K_r vs. $2b/L$ demonstrates that the present model gives reasonable predictions for the wave reflection of perforated/slotted walls.

3. Results and discussions

In this section, we present the numerical results and the comparisons of the present OWC-breakwater system and the conventional breakwater will be conducted. Firstly, the resonance frequency of an OWC caisson is examined using the validated model. The variation of Helmholtz resonance frequency of OWC as a function of chamber wall draft (d_2) is illustrated. Then, the influence of the perforated wall and the key parameters are examined to illustrate the wave absorption capability for long waves. Finally, comparisons will be presented and discussed.

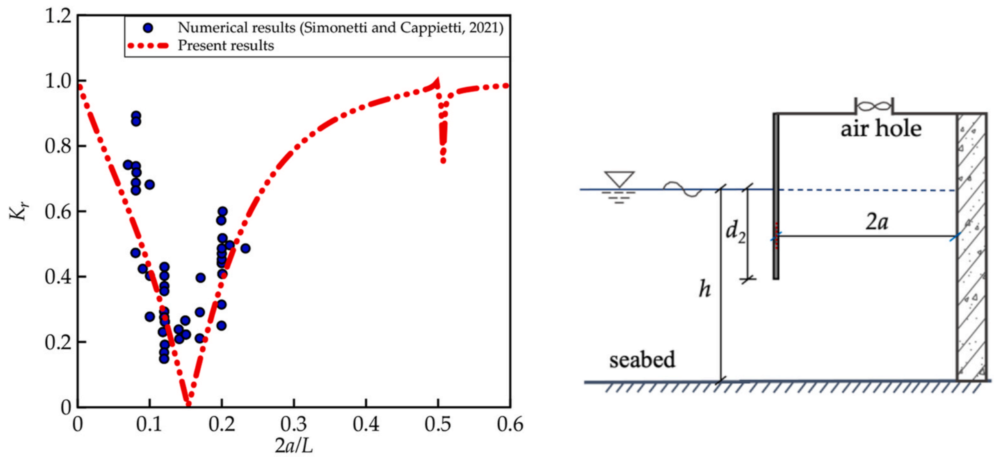
3.1. Effect of the chamber wall draft on the resonance frequency of OWC

In this subsection, we theoretically examine the resonance characteristic of an OWC. The geometrical and wave parameters are fixed as $d_2/h = 0.85$, $a/h = 0.25$, $l/h = 0.5$, $\theta = 0$, and $h = 10$m. Variations of reflection coefficient and hydrodynamic efficiency are shown in Fig. 6. It is found that the trend of reflection coefficient is opposite to that of hydrodynamic efficiency. The minimum reflection coefficient and the maximum hydrodynamic efficiency are observed at the resonance frequency ($kh = 0.78$), where the piston mode resonance of water column is triggered. Interestingly, at the resonance frequency, the wave reflection becomes zero due to fully absorbed wave energy by the OWC device. The qualified hydrodynamic efficiency/wave absorption capacity occurs at the neighborhood of the resonance frequency. The shortcoming of the single-chamber OWC is its narrow bandwidth for effective wave energy absorption.

Fig. 7 shows the variations of the resonance frequency of the OWC with the draft of the chamber wall (i.e., volume of the water column). It is evident that, within the present potential-flow based results, the resonance frequency decreases as the chamber wall approaches the seabed. While the wet opening, defined as the gap between the tip of the chamber wall and the seabed, of the OWC chamber is small (i.e., $(h - d_2)^2/(2ah) \sim o(1)$), the resonance of OWC is known as the Helmholtz resonance of the OWC. Namely, the full wave absorption at the lower resonance frequency ω_c is triggered when the wet opening is small. Note that, generally, ω_c located in lower frequency region leads to the long wave absorption of the Helmholtz resonance OWC.

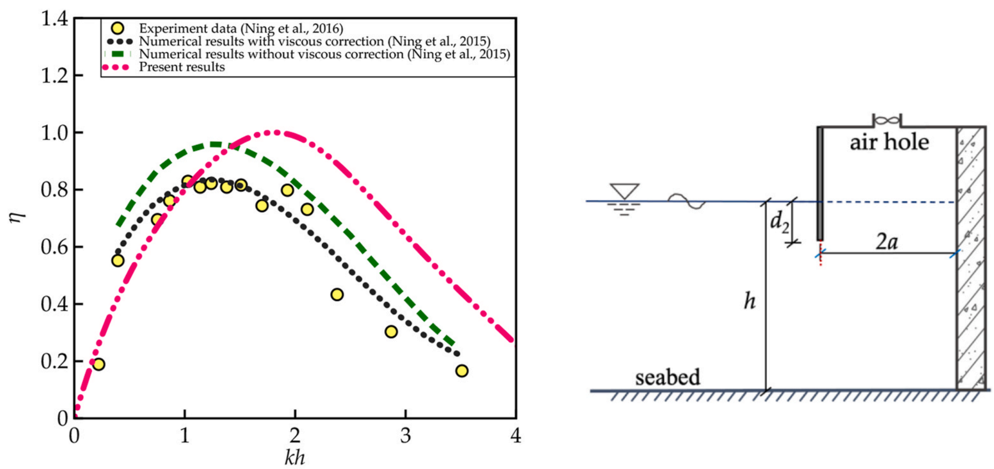
Table 2
Results of hydrodynamic efficiency η, reflection coefficient K_r, and $\eta + K_r^2$ for different porous coefficients G.

Wavenumber		$kh = 0.5$	$kh = 1$	$kh = 1.5$	$kh = 2$	$kh = 2.5$	$kh = 3$	$kh = 3.5$	$kh = 4$
G = ε (1+1i), ε = 1E-7	η	0.8630	0.9353	0.0024	0.0086	0.0127	0.0109	0.0068	0.0042
	K_r	0.3702	0.2543	0.9988	0.9957	0.9936	0.9945	0.9966	0.9979
	$\eta + K_r^2$	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
G = ε (1+1i), ε = 1E7	η	0.5650	0.7003	0.2330	0.0852	0.0314	0.0120	0.0057	0.0007
	K_r	0.6596	0.5475	0.8758	0.9565	0.9842	0.9940	0.9971	0.9997
	$\eta + K_r^2$	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000



(a) (b)

Fig. 3. (a) Comparisons of the reflection coefficient K_r of the present results and the numerical results in Simonetti and Cappietti (2021); (b) Sketch for key parameters of the OWC caisson.



(a) (b)

Fig. 4. (a) Comparisons of the hydrodynamic efficiency η of the present results with the experimental data in Ning et al. (2016) and numerical results in Ning et al. (2015). (b) Sketch for key parameters of the OWC caisson.

Besides, comparisons of the natural frequency predicted using the present potential flow solver and the formula suggested by Veer and Thorlen (2008) are plotted in Fig. 7. It is concluded that the general trend of the two aspects is similar. But the resonance frequency presented by Veer and Thorlen (2008) is greater than the present results over the whole frequency range with the exception of $kh = 0.15$. This is due to that the formula of $\omega_c = \sqrt{g / \left(d_2 + 0.41\sqrt{S}\right)}$ is derived from the moonpool problem, where S is the cross-sectional area of the moonpool.

To link the Helmholtz resonator for acoustic attenuation and the Helmholtz resonance-based OWC, we recall the principle of Helmholtz resonator. As shown in Fig. 8 (a), the Helmholtz resonator is a container of gas (usually air) with a small open hole or neck. Principally, the Helmholtz resonator works by resonating longer waves, which is known as Bass trap, or Helmholtz resonance (Jun et al., 2021). Bass trap is widely used to absorb low-frequency voice. The frequency of the Helmholtz resonance is determined by $f_c = \frac{\nu}{2\pi}\sqrt{S_c / (V_c(l_c + \delta_c))}$, where $\nu = 340$ m/s (see Fig. 8 (a)), and δ_c is the end correction (Ingard, 1953; Chanaud, 1994). It is known that the resonance frequency decreases with decreasing opening area. Since the fluid properties are different (i.e., air and water), Helmholtz resonance formula of $f_c = \frac{\nu}{2\pi}\sqrt{S_c / (V_c(l_c + \delta_c))}$ may not accurately be suitable for predicting the resonance frequency of the OWC. The gap between the seabed and the chamber wall of a Helmholtz resonance OWC acts as the neck of Helmholtz resonator, and is much smaller than the water depth. A similar explanation can be found in harbor resonance reported in Miles and Lee (1975). The OWC chamber surrounded by the chamber wall, partition wall, top wall, seawall, and seabed acts as the cavity of a Helmholtz resonator (i.e., Helmholtz resonance OWC). When the gap between the chamber wall and the seabed decreases, the resonance frequency of OWC decreases to a lower frequency regime. Thus, with the implementation of the PTO damping, the long wave absorption may be achieved by properly designing the OWC chamber. Physically, the Helmholtz resonator for acoustic attenuation and the Helmholtz resonance-based OWC in Fig. 8 work in a similar principle with the unique capability of long wave absorption.

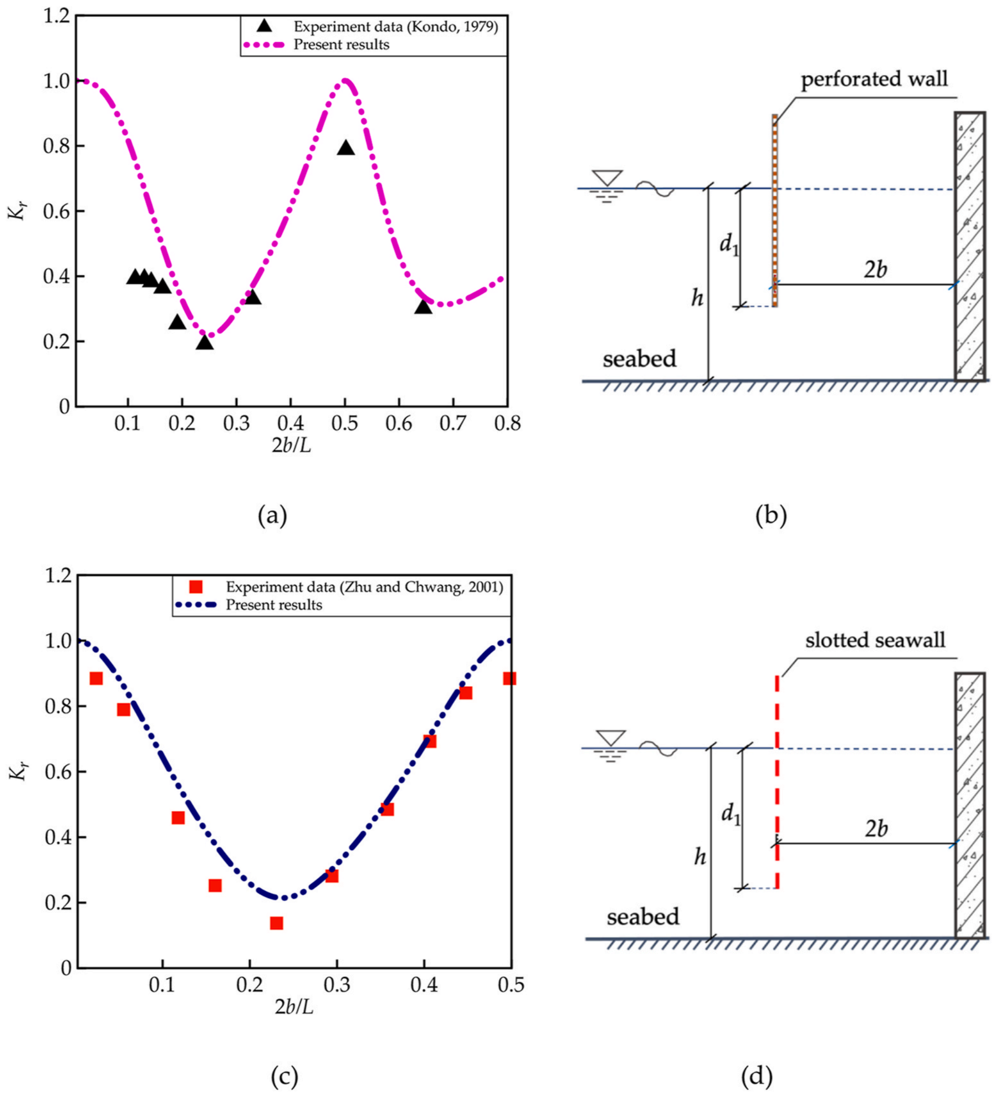


Fig. 5. (a) Comparisons of the reflection coefficient K_r of the present results and the published data for perforated wall. (b) Geometrical sketch of perforated wall. The physical and geometrical parameters are $\gamma = 0.2$, $f = 9.0$, $\delta/h = 0.012$, $C_m = 0$, $d_1/h = 0.99$, $2b/h = 1.0$, $\theta = 0$, and $h = 0.5$m. (c) Comparisons of the reflection coefficient K_r between the present results and the published data for slotted seawall. (d) Geometrical sketch of slotted seawall. The physical and geometrical parameters are $\gamma = 0.2$, $f = 9.5$, $\delta/h = 0.0094$, $C_m = 0$, $d_1/h = 0.99$, $kh = 2.08$, $\theta = 0$, and $h = 0.32$m.

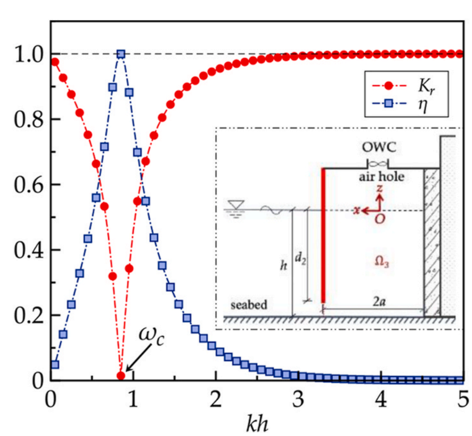


Fig. 6. Results of the reflection coefficient K_r and the hydrodynamic efficiency η. The geometrical and wave parameters are $d_2/h = 0.85$, $a/h = 0.25$, $l/h = 0.5$, $\theta = 0$, and $h = 10$m.

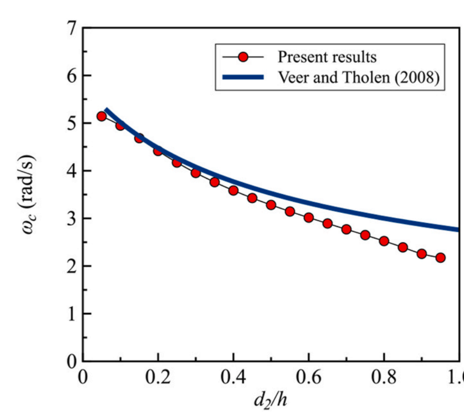


Fig. 7. Variations of the resonance frequency ω_c of water column against the draft of the chamber wall d_2. The geometrical and wave parameters are $a/h = 0.25$, $l/h = 0.5$, $\theta = 0$, and $h = 10$m.

3.2. Effect of the perforated wall

As discussed above, the frequency bandwidth of the single-chamber OWC is narrow for effective wave absorption. Several studies on conventional single-chamber OWC wave energy devices confirmed peak hydrodynamic efficiency at natural resonance frequency (Falcão, 2002; Marjani et al., 2008; Drew et al., 2009; Iturrioz et al., 2015). Zhang et al. (2021) found that the perforated plate may act as an effective wave absorber at the higher frequency range and broaden the frequency

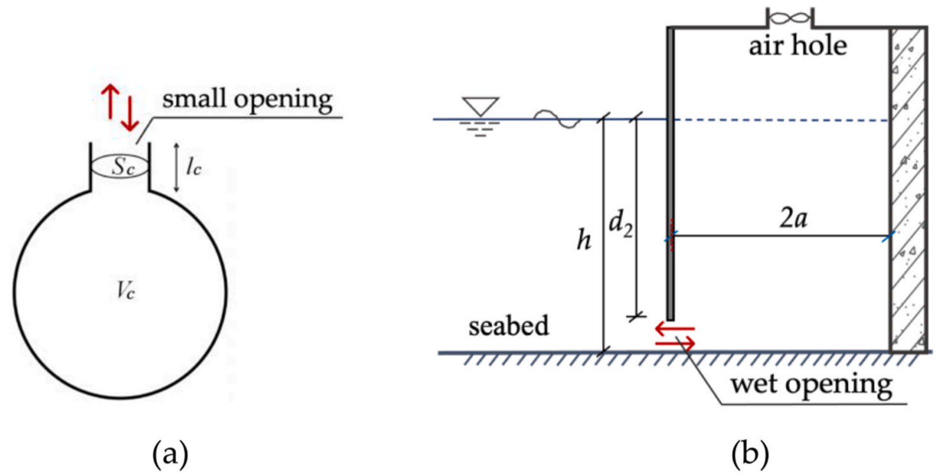


Fig. 8. Sketch of (a) a Helmholtz resonator for acoustic attenuation. V_c, S_c, l_c are chamber volume, cross-sectional area of the opening, and the length of the opening, respectively; (b) Helmholtz-resonance OWC device.

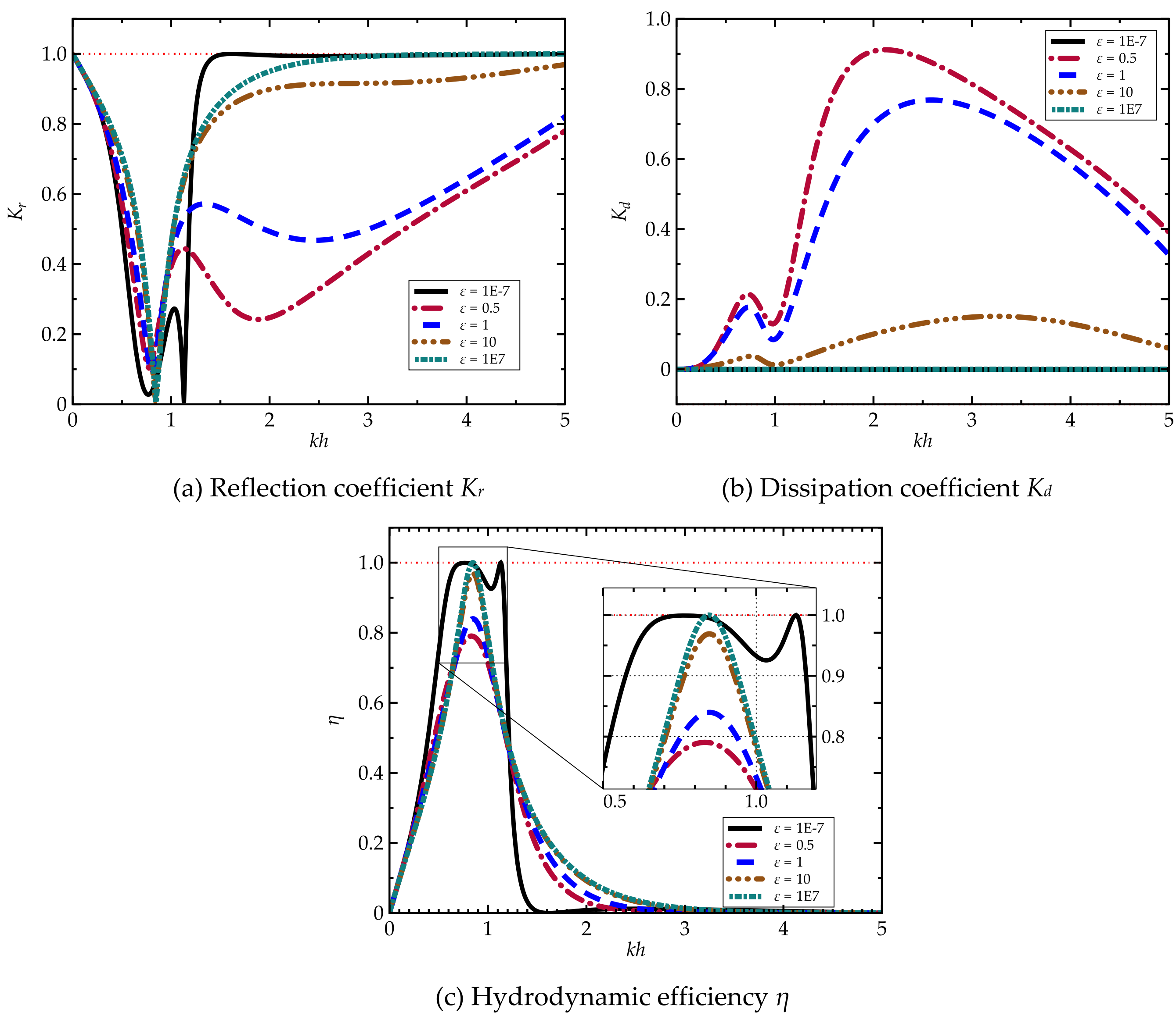


Fig. 9. (a) Reflection coefficient K_r, (b) Dissipation coefficient K_d, and (c) hydrodynamic efficiency η against kh for different porous coefficient $G = \varepsilon$ (1+1i). The geometrical and wave parameters are $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $l/h = 0.5$, $\theta = 0$, and $h = 10$m.

bandwidth of OWC caisson for wave attenuation. In this subsection, the performance of Helmholtz resonance-based OWC caisson array equipped with a perforated wall is examined with the anticipation of wave absorption over a broader frequency range.

The variations of the reflection coefficient, dissipation coefficient, and the hydrodynamic efficiency against kh are illustrated in Fig. 9. Note that the dissipation coefficient represents the energy dissipated by the perforated plate. The hydrodynamic efficiency represents the wave energy extraction efficiency by the OWC device. The present OWC-breakwater system absorbs wave energy by the combination of wave energy dissipation of perforated wall and wave energy extraction of Helmholtz resonance OWC device. The geometrical and wave parameters are fixed as $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $l/h = 0.5$, $\theta = 0$, $h = 10$m, $G = \varepsilon$ (1+1i) with ε = 1E-7, 0.5, 1, 10 and 1E7.

Fig. 9 (a) indicates that the porous coefficient G has significant effects on the reflection coefficient over the medium-frequency range of $1 < kh < 5$, and that only at a proper porous coefficient (i.e., $G = 0.5$ (1+1i)), a broader frequency bandwidth with effective wave absorption can be achieved. The dimensionless free surface elevation at $kh = 0.7$ corresponding to the minimum reflection coefficient is shown in Fig. 10. The significant mitigation of the wave reflection at $1 < kh < 5$ is due to the energy dissipation by the perforated wall. As is shown in Fig. 9 (b), a peak dissipation coefficient occurs at the medium-frequency range, corresponding to the significant reduction of reflection coefficient in Fig. 9 (a).

A minimum reflection coefficient is observed at a low frequency corresponding to the resonance frequency of the water column. From Fig. 9 (c), we observe a peak η in the low frequency range. The hydrodynamic characteristics of the water column change slightly as the porous coefficient of the perforated wall varies. Two peaks of hydrodynamic efficiency (i.e., two valleys of K_r) are observed for $\varepsilon = 0$ due to the fact that multiple resonance of the water column may be triggered for multi-chamber OWC device (Ning et al., 2018; Zheng et al., 2020; Zhao et al., 2021b).

From Fig. 6, theoretical results indicate that minimum reflection coefficient and peak hydrodynamic efficiency occurs at the low frequency region of $0.7 < kh < 0.9$, close to the piston resonance of OWC at $kh = 0.78$ when the wet opening is small. This leads to the full absorption of long waves. The Helmholtz resonance OWC device performs well

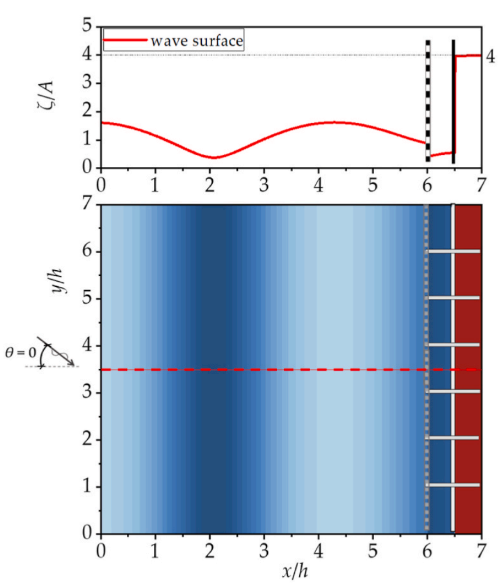


Fig. 10. (Lower) top view of wave amplitude distribution for $kh = 0.7$ and (Upper) side view of wave amplitude ($y/h = 3.5$, $0 < x/h < 7$). The geometrical and wave parameters are $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $l/h = 0.5$, $\theta = 0$, $G = 0.5$ (1+1i), and $h = 10$m.

in long wave absorption over a narrow frequency range. The perforated wall and Helmholtz resonance OWC dissipate and extract wave energy at different frequency ranges as illustrated in Fig. 9 (b) and (c). It follows that the perforated wall with proper porosity increases the frequency bandwidth for effective wave absorption.

3.3. Effect of the incident wave angle

The performance of the coastal structures under the action of oblique waves is important (Gent and Werf, 2019; Jalón et al., 2019; Gent, 2021). In this section, we investigate the influence of incident wave angle on Helmholtz resonance OWC caisson array with a perforated wall. The geometrical and wave parameters are fixed as $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $l/h = 0.5$, $G = 0.5$ (1+1i), $h = 10$m, $\theta = 0$, $\pi/6$, $\pi/4$, $\pi/3$, and $5\pi/12$.

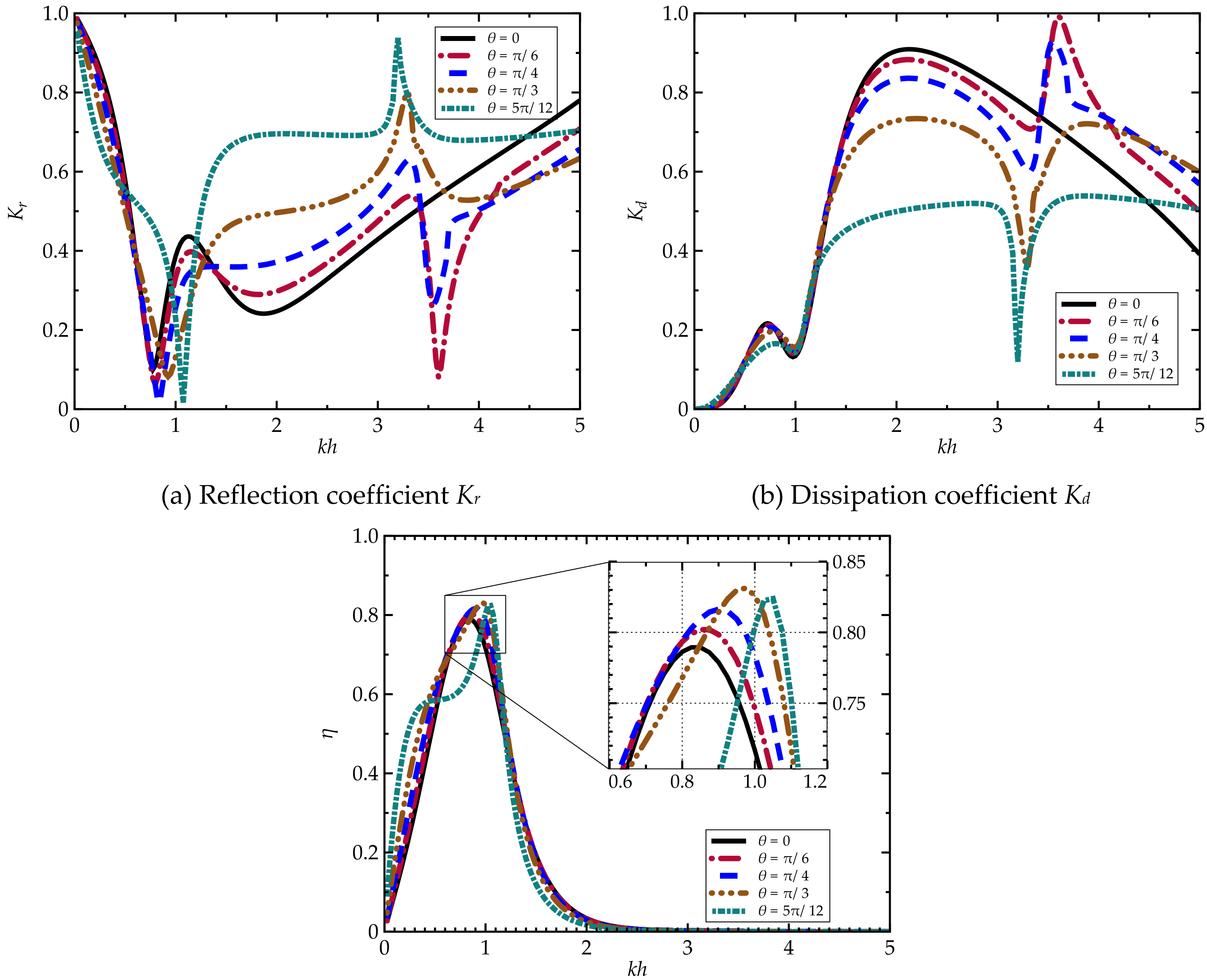
As shown in Fig. 11, the incident wave angle affects the dissipation coefficient significantly but modifies the hydrodynamic efficiency slightly. There exists a critical frequency $k_ch \in (3, 3.5)$, beyond which the dissipation coefficient increases with increasing wave angle. However, at the lower-frequency range $k < k_c$, the dissipation coefficient decreases with increasing wave angle. To distinguish the wavefield at different wave frequencies, Fig. 12 (a) and (b) show the wave amplitude distributions corresponding to different wave frequencies (i.e., kh = 1.075 and 3.205 with $\theta = 5\pi/12$). Interestingly, it is observed that the piston resonance of OWC is triggered at a low frequency of $kh = 1.075$ (shown in Fig. 12 (a)), which leads to the minimum wave reflection and peak wave absorption. In addition, the sloshing resonance of water column in the along-shore direction can be observed at $kh = 3.205$, which corresponds to the peak wave reflection (see Fig. 11 (a)). This is not beneficial for coastal protection and wave power extraction.

By comparing Fig. 11 (a) and (b), we found that the influence of the incident wave angle on the dissipation coefficient is opposite to that of the reflection coefficient. Specifically, the reflection coefficient increases with increasing wave angle for $1.5 < kh < 3$. Notably, the incident wave angle slightly affects the minimum reflection coefficient's location. Accordingly, the peak hydrodynamic efficiency and its location change slightly as well. It follows that the wave absorption performance of the system remains strong for long waves for various incident wave angles.

3.4. Effect of OWC length in the along-shore direction

In this subsection, we examine the influence of OWC length in the along-shore direction on the performance of Helmholtz resonance OWC caisson array. The reflection coefficient, dissipation coefficient, and hydrodynamic efficiency for different OWC length l/h are shown in Fig. 13. The geometrical and wave parameters are fixed as $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $G = 0.5$ (1 + 1i), $h = 10$m, $\theta = \pi/4$, $l/h = 0.3$, 0.4, 0.5, 0.6 and 0.7.

It is evident from Fig. 13 that OWC length in the along-shore direction affects the reflection and dissipation coefficient considerably. However, the hydrodynamic efficiency is only slightly modified as the OWC chamber length varies in oblique waves. The critical frequency k_c, where the reflection and dissipation coefficient changes dramatically, shifts to lower-frequency range as the OWC length in the along-shore direction increases. This is because the characterized wavelength that triggers the sloshing resonance increases with increasing l/h. But the changes of OWC length slightly affect the reflection and dissipation coefficient for the frequency range of $k < k_c$. The wave reflection is satisfactory at lower-frequency range and the significant dissipation coefficient occurs at high frequency range. It follows that the absorption capability of long waves is not affected by the OWC length according to the present results. Consequently, the long wave absorption capacity of the proposed OWC-breakwater system remains effective for a range of OWC lengths in the along-shore direction.



(a) Reflection coefficient K_r

(b) Dissipation coefficient K_d

(c) Hydrodynamic efficiency η

Fig. 11. (a) Reflection coefficient K_r, (b) Dissipation coefficient K_d, and (c) hydrodynamic efficiency η against kh for incident wave angle $\theta = 0$ to $5\pi/12$. The geometrical and wave parameters are $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $l/h = 0.5$, $G = 0.5(1+1i)$, and $h = 10$m.

4. Comparisons with conventional breakwaters and further discussions

Rubble mound breakwaters and perforated wall breakwaters are widely used for coastal protection. In this section, we compare the wave reflection of the present OWC-breakwater system with that of the classical bottom-mounted rubble mound breakwater, single- and multi-perforated wall breakwater.

Fig. 14 shows the comparisons of the reflection coefficient of the present OWC-breakwater system and the sloping rubble mound breakwater. The corresponding analytical results and the experimental data are documented in Liu and Faraci (2014) and Pérez-Romero et al. (2009), respectively. The wave and geometrical parameters are fixed as $h = 0.24$m, $d_1/h = 0.75$, $d_2/h = 0.85$, $a/h = b/h = 0.25$, $G = 0.3$ (1+1i) and $\theta = 0$. The width of the OWC chamber $2a$ in the present study is the same as the breadth of the rubble mound caisson B in Liu and Faraci (2014). Fig. 14 shows that the reflection coefficient of the present breakwater system is much smaller than that of the sloping rubble mound breakwater at lower-frequency range (i.e., $B/L < 0.1$). A collapse of reflection coefficient is observed at $B/L = 0.056$, due to the Helmholtz resonance of the OWC. These comparisons demonstrate that the longer waves absorption capacity of the present breakwater system is superior compared to that of the conventional rubble mound breakwaters.

Fig. 15 shows the comparisons of the predicted K_r for the present breakwater system and the experimental results of Lee and Shin (2014) for a single perforated wall caisson. Here we consider two cases with $G = \infty$ and $G = 0.15$ (1+1i). The other parameters are fixed as $h = 0.5$ m, $2a/h = 0.6$, $2b/h = 0.3$, $d_1/h = 0.7$, $d_2/h = 0.85$, and $\theta = 0$. The width of the OWC chamber, $2a$, in the present study is equivalent to the breadth of the partially perforated wall caisson B in Lee and Shin (2014). It is evident from Fig. 15 that the present OWC-breakwater system shows better long wave absorption at a low frequency range ($0.05 < B/L < 0.1$) than the conventional single perforated wall caisson. When the incident wave frequency is close to the natural frequency of Helmholtz resonance OWC, the long wave energy is captured by the oscillating water column rather than dissipated by the perforated wall. In addition, the porous wall is efficient in dissipating medium waves, the present Helmholtz resonance OWC array equipped with a perforated wall shows a broader frequency bandwidth of wave attenuation for both medium and long waves.

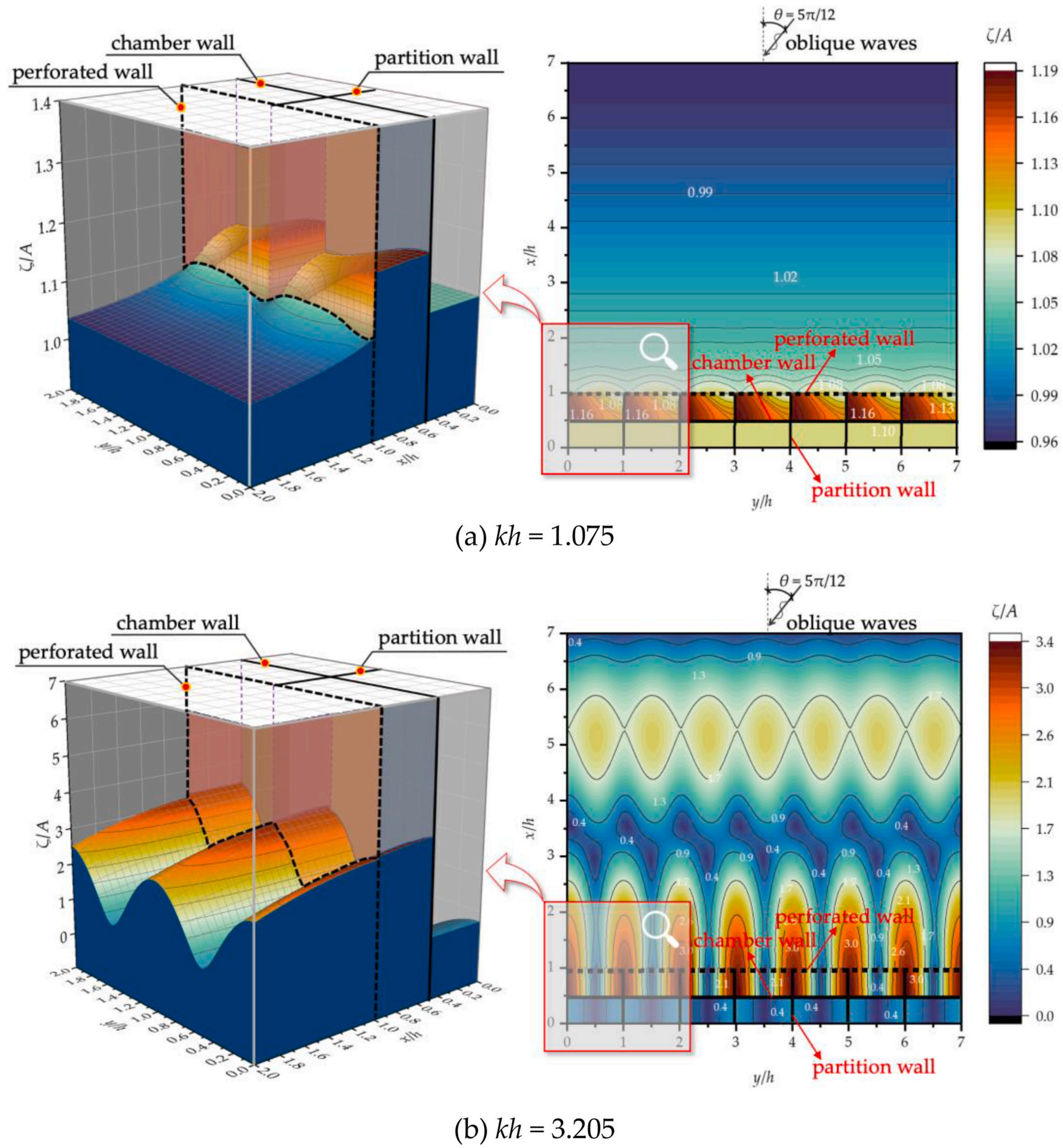


Fig. 12. Three-dimensional wave amplitude distributions at (a) $kh = 1.075$ and (b) $kh = 3.205$ for $\theta = 5\pi/12$. The geometrical and wave parameters are $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $l/h = 0.5$, $G = 0.5(1+1i)$, and $h = 10$m.

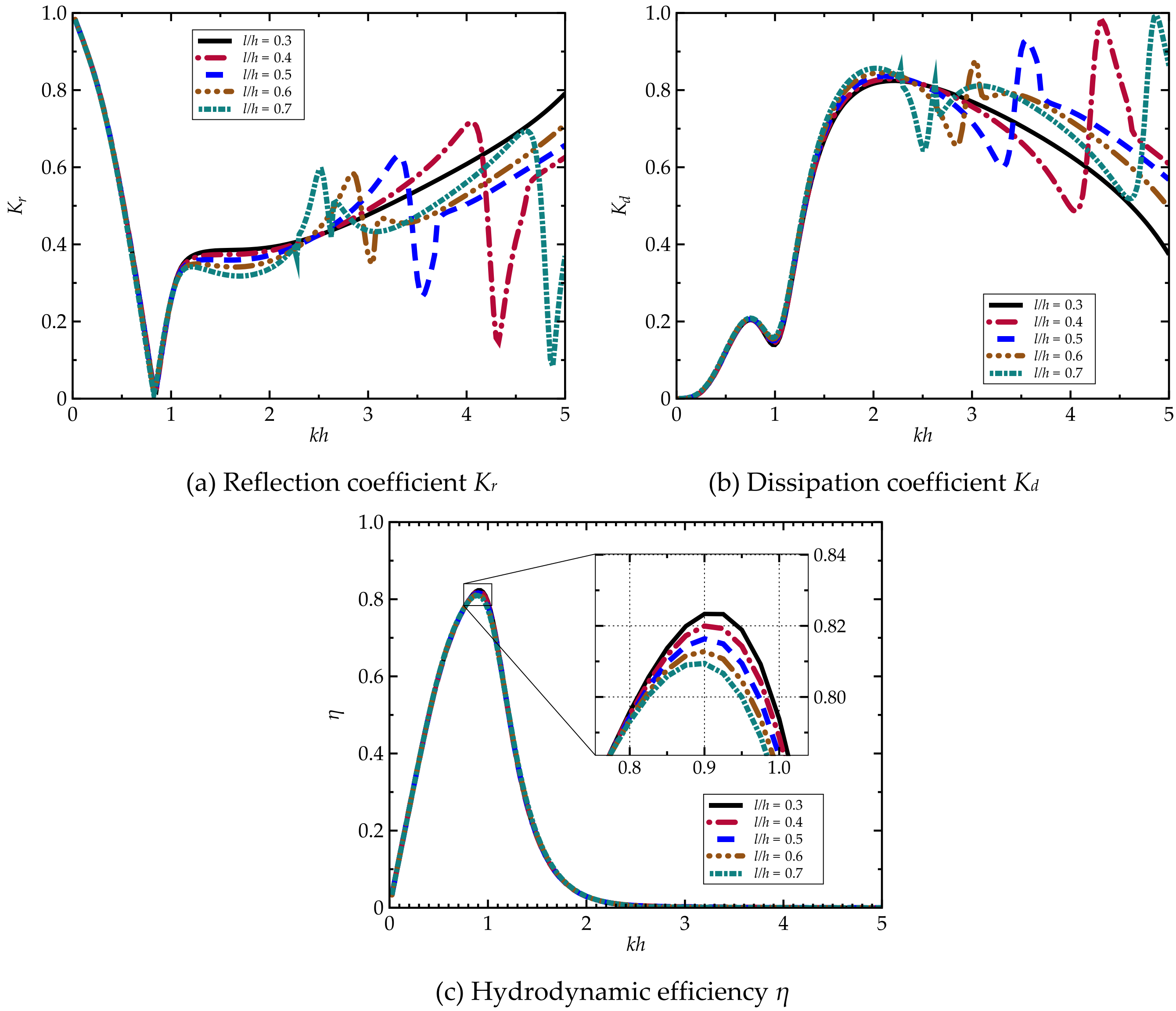


Fig. 13. (a) Reflection coefficient K_r, (b) dissipation coefficient K_d, and (c) hydrodynamic efficiency η against kh for the OWC chamber length in the along-shore direction $l/h = 0.3$ to 0.7. The geometrical and wave parameters are $d_1/h = d_2/h = 0.85$, $a/h = b/h = 0.25$, $G = 0.5\ (1{+}1\mathrm{i})$, $h = 10\mathrm{m}$, and $\theta = \pi/4$.

Liu et al. (2012) conducted thorough theoretical investigations on a coastal breakwater consisting of multiple perforated walls. Here we compared the reflection coefficient of the present results and Liu et al.'s (2012) results. The premise is that the total breadth of the present breakwater is the same as that of the triple-chamber perforated caisson. From the results shown in Fig. 16, we found that present reflection coefficient is obviously lower than that of Liu et al. (2012) at the lower-frequency range. These theoretical comparisons confirmed the advantage of the wave absorption capacity in lower-frequency range.

Overall, the comparisons in Fig. 14–16 suggest that the wave absorption performance of the Helmholtz resonant OWC with a perforated wall is much better than that of the classical dissipative type breakwaters at the lower-frequency range. Basic rules and the beneath mechanism of the present OWC-breakwater system are illustrated theoretically. To the best of author's knowledge, there is a lack of numerical/experimental investigation of the Helmholtz resonance of OWC which is a worthwhile future work.

5. Conclusions

Long wave absorption is of interest to coastal engineers and has not been solved well until now. In this paper, the absorption of long waves by a novel Helmholtz resonance OWC caisson array with a perforated front wall is investigated. An efficient semi-analytical model for wave interaction with the Helmholtz resonance OWC caisson array was developed and the velocity singularity at the tip of both the perforated wall and the thin chamber wall was resolved mathematically. The developed semi-analytical tool is validated against numerical simulation and experimental data. The natural frequency of the water column shifts to the lower-frequency range as the chamber wall approaches the seabed, which is beneficial for extracting long waves energy. The wave absorption performance and mechanism of the present OWC-breakwater system, especially for medium and long waves, is confirmed theoretically. Parametrical investigations are conducted for an extensive range of geometrical and wave parameters to observe fundamental rules

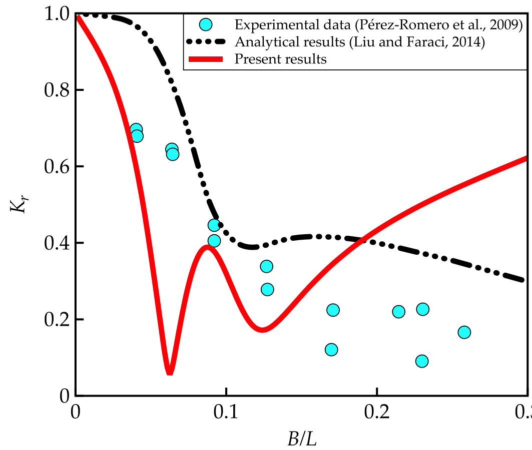


Fig. 14. Comparisons of the wave reflection of the Helmholtz-resonance OWC with perforated wall and that of the sloping rubble mound breakwater. The geometrical and wave parameters are: water depth $h = 0.24$m, perforated wall draft $d_1/h = 0.75$, OWC chamber wall draft $d_2/h = 0.85$, distance between the perforated wall and the OWC chamber $2b/h = 0.5$, breadth of the OWC chamber $2a/h = 0.5$, porous coefficient of the perforated wall $G = 0.3$ (1+1i), for incident wave angle $\theta = 0$.

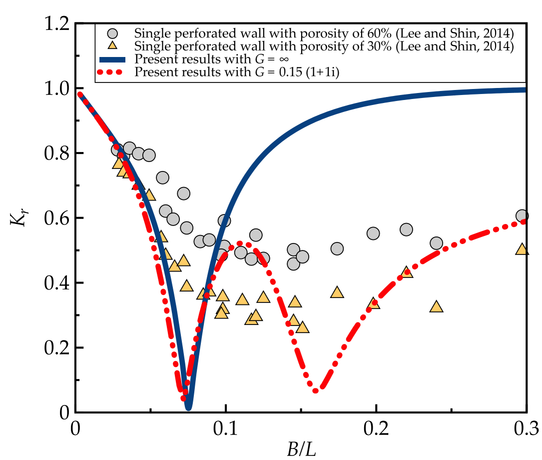


Fig. 15. Comparisons of the predicted wave reflection K_r by the present analytical model for the Helmholtz resonance OWC with and without the perforated wall and the experimental results for perforated wall caissons with the porosity of 60% and 30% in Lee and Shin (2014). The geometrical and wave parameters are: water depth $h = 0.5$m, distance between the perforated wall and the OWC chamber $2b/h = 0.3$, breadth of the OWC chamber $2a/h = 0.6$, perforated wall draft $d_1/h = 0.7$, OWC chamber wall draft $d_2/h = 0.85$, the porous coefficient of the perforated wall $G = 0.15(1+1i)$, for normal incident wave angle $\theta = 0$.

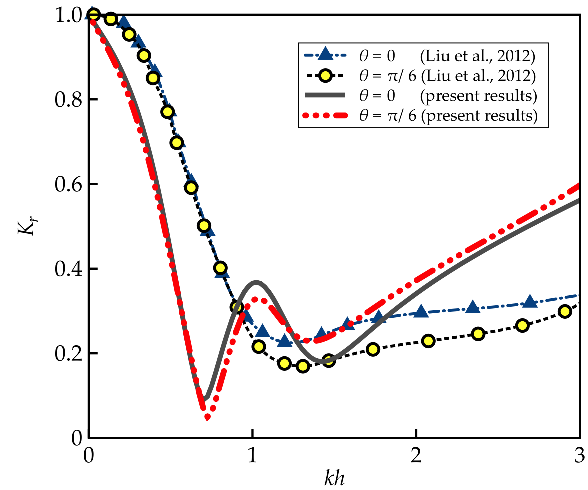


Fig. 16. Comparisons of the present analytical results for a Helmholtz resonance OWC with a perforated wall and that of Liu et al. (2012) for triple-chamber perforated caisson. The geometrical and wave parameters are: water depth $h = 10$m, perforated wall draft $d_1/h = 0.85$, OWC chamber wall draft $d_2/h = 0.85$, distance between the perforated wall and the OWC chamber $2b/h = 0.6$, breadth of the OWC chamber $2a/h = 0.6$, porous coefficient of the perforated wall $G = 0.4(1+1i)$, for incident wave angle $\theta = 0$ and $\pi/6$.

associated with the hybrid system. The incident wave angle, draft of OWC chamber wall, along-shore width of each unit, and the porous coefficient of the perforated wall are identified as the key factors that affect the performance of the OWC-breakwater system. Moreover, comparisons with classical coastal breakwaters (i.e., multi-chamber perforated walls and rubble mound breakwaters) were made to illustrate the unique features of the present OWC-breakwater system. The following conclusions can be drawn.

1) We developed an efficient semi-analytical solution for solving the problem of water wave interaction with OWC array with perforated wall. The velocity potential solution is verified using Haskind relation and wave energy conservation rule. In addition, the semi-analytical model is in good agreement with the published experimental and numerical results, therefore, can be used to predict the performance of Helmholtz resonance OWC caisson array with perforated wall.
2) Effective wave absorption with a broader frequency bandwidth can be achieved by combining the Helmholtz resonance OWC device and the perforated wall. Over the hybrid system, the incident wave energy is transformed into 3 parts: (a) the dissipated energy by the perforated wall, (b) the extracted energy by Helmholtz resonance OWC caisson, and (c) the wave energy reflected to the sea. Within the investigated frequency range, most of the incident wave energy was reflected the sea at a very low frequency ($kh < 0.5$). At the relatively low frequency ($0.5 < kh < 1$), however, the majority of the incident wave energy is extracted by the Helmholtz resonance OWC caisson. At the higher frequency ($kh > 1$), the incident wave energy is mainly dissipated by the perforated wall.
3) The length of each OWC caisson unit in the along-shore direction affects the wave resonance in the along-shore direction, therefore, it is a key contributing factor to the hydrodynamic characteristic of the present OWC-breakwater system. The present study indicates that the along-shore chamber length of each Helmholtz resonance OWC unit has little effect on the long wave absorption but significant effect on wave reflection of the OWC caisson array due to the sloshing motion inside the chamber.
4) Compared with the conventional breakwaters (i.e., multi-chamber perforated walls and rubble mound breakwaters), the present OWC-breakwater system with a perforated wall has better wave absorption capability for long waves, due to the Helmholtz resonance of the OWC at the long wave frequency. This novel hybrid scheme can be installed to compensate for the deficiency of long wave absorption by conventional breakwaters.

CRediT authorship contribution statement

Xuanlie Zhao: Methodology, Investigation, Formal analysis, Funding acquisition, Writing – original draft. **Yang Li:** Methodology, Investigation, Formal analysis, Validation, Writing – original draft. **Qingping Zou:** Discussion, Writing – review & editing. **Duanfeng Han:** Funding acquisition, Project administration. **Jing Geng:** Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The expressions of diffraction potential are shown as:

$$\varphi_{\mathrm{D}}^{(1)} = -\frac{\mathrm{i}gA}{\omega}\sum_{m=-\infty}^{\infty} Y_m(y)\sum_{n=1}^{\infty} A_{mn}^{(1)} \mathrm{e}^{p_{mn}(x-a-2b)} Z_n(z), \tag{A1 a}$$

$$\varphi_{\mathrm{D}}^{(2)} = -\frac{\mathrm{i}gA}{\omega}\sum_{m=1}^{\infty} \overline{Y}_m(y)\sum_{n=1}^{\infty}\left[A_{mn}^{(2)} \mathrm{e}^{-\overline{p}_{mn}(x-a-2b)} + A_{mn}^{(3)} \mathrm{e}^{\overline{p}_{mn}(x-a)}\right] Z_n(z), \tag{A1 b}$$

and

$$\varphi_{\mathrm{D}}^{(3)} = -\frac{\mathrm{i}gA}{\omega}\sum_{m=1}^{\infty} \overline{Y}_m(y)\sum_{n=1}^{\infty}\left[A_{mn}^{(4)} \mathrm{e}^{-\overline{p}_{mn}(x-a)} + A_{mn}^{(5)} \mathrm{e}^{\overline{p}_{mn}(x+a)}\right] Z_n(z), A_{mn}^{(5)} = A_{mn}^{(4)} \mathrm{e}^{2\overline{p}_{mn}a}, \tag{A1 c}$$

where $A_{mn}^{(1)}$, $A_{mn}^{(2)}$, $A_{mn}^{(3)}$, $A_{mn}^{(4)}$ and $A_{mn}^{(5)}$ are unknown coefficients of diffraction potential. Please note that, by substituting the velocity potential $\varphi_{\chi}^{(3)}$ into the rigid boundary conditions of $\frac{\partial \varphi_{\chi}^{(j)}}{\partial x} = 0,\ x = -a,\ z \in [-h, 0],\quad y \in [-l, l],\ j = 3$, in Eq. (8), we can derive the relation of $A_{mn}^{(5)} = A_{mn}^{(4)} \mathrm{e}^{2\overline{p}_{mn}a}$. The y-direction eigenfunctions of $Y_m(y)$ in domain Ω_1 and $\overline{Y}_m(y)$ in domain Ω_j ($j = 2$ and 3) are expressed as

$$Y_m(y) = \mathrm{e}^{\mathrm{i}\gamma_m y} \ \text{ with } \ \gamma_m = k_y + \frac{2\pi m}{2l}, m = 0, \pm 1, \pm 2, \ldots, \tag{A2 a}$$

and

$$\overline{Y}_m(y) = \cos[\overline{\gamma}_m(l-y)] \ \text{ with } \ \overline{\gamma}_m = \frac{(m-1)\pi}{2l}, m = 1, 2, 3, \ldots, \tag{A2 b}$$

respectively. $Y_m(y)$ and $\overline{Y}_m(y)$ satisfy the orthogonal relation with respect to $y \in [-l, l]$, which can be written as

$$\int_{-l}^{l} Y_m(y) Y_v^*(y)\mathrm{d}y = \begin{cases} 2l, & m = v, \\ 0, & m \neq v, \end{cases} \tag{A3 a}$$

and

$$\int_{-l}^{l} \overline{Y}_m(y)\overline{Y}_v(y)\mathrm{d}y = \begin{cases} 2l, & m = v = 1, \\ l, & m = v \neq 1, \\ 0, & m \neq v, \end{cases} \tag{A3 b}$$

respectively. The z-direction eigenfunction of $Z_n(z)$ is expressed as

$$Z_n(z) = \frac{\cos[\kappa_n(z+h)]}{\cos(\kappa_n h)}. \tag{A4}$$

$Z_n(z)$ satisfies the orthogonal relation with respect to $z \in [-h, 0]$, which can be written as

$$\int_{-h}^{0} Z_n(z) Z_u(z)\mathrm{d}z = \begin{cases} \dfrac{1}{\cos^2 \kappa_n h}\left(\dfrac{h}{2} + \dfrac{\sin 2\kappa_n h}{4\kappa_n}\right), & n = u, \\ 0, & n \neq u. \end{cases} \tag{A5}$$

The expressions of radiation potential are shown as

$$\varphi_{\mathrm{R}}^{(1)} = -\frac{\mathrm{i}gA}{\omega}\sum_{m=-\infty}^{\infty} Y_m(y)\sum_{n=1}^{\infty} \widehat{A}_{mn}^{(1)} \mathrm{e}^{p_{mn}(x-a-2b)} Z_n(z), \tag{A6 a}$$

$$\varphi_{\mathrm{R}}^{(2)} = -\frac{\mathrm{i}gA}{\omega}\sum_{m=1}^{\infty}\overline{Y}_m(y)\sum_{n=1}^{\infty}\left[\widehat{A}_{mn}^{(2)}\mathrm{e}^{-\overline{p}_{mn}(x-a-2b)} + \widehat{A}_{mn}^{(3)}\mathrm{e}^{\overline{p}_{mn}(x-a)}\right]Z_n(z), \tag{A6 b}$$

and

$$\varphi_{\mathrm{R}}^{(3)} = -\frac{\mathrm{i}gA}{\omega}\sum_{m=1}^{\infty}\overline{Y}_m(y)\sum_{n=1}^{\infty}\left[\widehat{A}_{mn}^{(4)}\mathrm{e}^{-\overline{p}_{mn}(x-a)} + \widehat{A}_{mn}^{(5)}\mathrm{e}^{\overline{p}_{mn}(x+a)}\right]Z_n(z) - \frac{\mathrm{i}}{\rho\omega}, \widehat{A}_{mn}^{(5)} = \widehat{A}_{mn}^{(4)}\mathrm{e}^{2\overline{p}_{mn}a}, \tag{A6 c}$$

where $\widehat{A}_{mn}^{(1)}$, $\widehat{A}_{mn}^{(2)}$, $\widehat{A}_{mn}^{(3)}$, $\widehat{A}_{mn}^{(4)}$ and $\widehat{A}_{mn}^{(5)}$ are unknown coefficients of radiation potential to be solved.

The coefficients of p_{mn} and $\overline{p}_{mn}$ are calculated by:

$$p_{mn} = \begin{cases} -\sqrt{\lambda_{m1}}, & \lambda_{m1} \geq 0,\ m = 0,\ \pm1,\ \pm2,\ \ldots,\ n = 1, \\ \mathrm{i}\sqrt{-\lambda_{m1}}, & \lambda_{m1} < 0,\ m = 0,\ \pm1,\ \pm2,\ \ldots,\ n = 1, \\ -\sqrt{\lambda_{mn}}, & m = 0,\ \pm1,\ \pm2,\ \ldots,\ n = 2,\ 3,\ \ldots, \end{cases} \tag{A7 a}$$

and

$$\overline{p}_{mn} = \begin{cases} \sqrt{\overline{\lambda}_{m1}}, & \overline{\lambda}_{m1} \geq 0,\ m = 1,\ 2,\ \ldots,\ n = 1, \\ \mathrm{i}\sqrt{-\overline{\lambda}_{m1}}, & \overline{\lambda}_{m1} < 0,\ m = 1,\ 2,\ \ldots,\ n = 1, \\ \sqrt{\overline{\lambda}_{mn}}, & m = 1,\ 2,\ \ldots,\ n = 2, 3, \ldots, \end{cases} \tag{A7 b}$$

where $\lambda_{mn} = \gamma_m^2 + \kappa_n^2$ and $\overline{\lambda}_{mn} = \overline{\gamma}_m^2 + \kappa_n^2$.

Considering diffracted waves and radiated waves, velocity potential characterizing the reflected waves at far field can be expressed as

$$\varphi_r = -\frac{\mathrm{i}gA}{\omega}\left[\sum_{m=-M_1}^{M_2}\mathrm{e}^{\mathrm{i}k\sin\theta_m y}A_{m1}^{(1)}\mathrm{e}^{\mathrm{i}k\cos\theta_m(x-a-2b)}Z_1(z) + p\sum_{m=-M_1}^{M_2}X_m(y)\widehat{A}_{m1}^{(1)}\mathrm{e}^{\mathrm{i}k\cos\theta_m(x-a-2b)}Z_1(z)\right]. \tag{A8}$$

By truncating the velocity potential expressions as finite series (i.e., the upper bound of m, n, s_1, and s_2 are selected as M, N, S_1, and S_2, respectively), a set of linear equations corresponding to the diffraction and radiation problem are expressed as

$$[\Xi]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times[(2M+1)N+M(3N+S_1+S_2+2)]}[X]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times1} = [B]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times1}, \tag{A9 a}$$

and

$$[\Xi]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times[(2M+1)N+M(3N+S_1+S_2+2)]}[\widehat{X}]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times1} = [\widehat{B}]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times1}, \tag{A9 b}$$

where the matrix of $[\Xi]$ can be described as

$$[\Xi]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times[(2M+1)N+M(3N+S_1+S_2+2)]}$$
$$= \begin{bmatrix} [\Xi]^{1,1}_{MN\times(2M+1)N} & [\Xi]^{1,2}_{MN\times MN} & [\Xi]^{1,3}_{MN\times MN} & O_{MN\times MN} & [\Xi]^{1,5}_{MN\times M(S_1+1)} & O_{MN\times M(S_2+1)} \\ [\Xi]^{2,1}_{(2M+1)N\times(2M+1)N} & [\Xi]^{2,2}_{(2M+1)N\times MN} & [\Xi]^{2,3}_{(2M+1)N\times MN} & O_{(2M+1)N\times MN} & O_{(2M+1)N\times M(S_1+1)} & O_{(2M+1)N\times M(S_2+1)} \\ [\Xi]^{3,1}_{M(S_1+1)\times(2M+1)N} & [\Xi]^{3,2}_{M(S_1+1)\times MN} & [\Xi]^{3,3}_{M(S_1+1)\times MN} & O_{M(S_1+1)\times MN} & O_{M(S_1+1)\times M(S_1+1)} & O_{M(S_1+1)\times M(S_2+1)} \\ O_{M(S_2+1)\times(2M+1)N} & [\Xi]^{4,2}_{M(S_2+1)\times MN} & [\Xi]^{4,3}_{M(S_2+1)\times MN} & [\Xi]^{4,4}_{M(S_2+1)\times MN} & O_{M(S_2+1)\times M(S_1+1)} & O_{M(S_2+1)\times M(S_2+1)} \\ O_{MN\times(2M+1)N} & [\Xi]^{5,2}_{MN\times MN} & [\Xi]^{5,3}_{MN\times MN} & O_{MN\times MN} & O_{MN\times M(S_1+1)} & [\Xi]^{5,6}_{MN\times M(S_2+1)} \\ O_{MN\times(2M+1)N} & O_{MN\times MN} & O_{MN\times MN} & [\Xi]^{6,4}_{MN\times MN} & O_{MN\times M(S_1+1)} & [\Xi]^{6,6}_{MN\times M(S_2+1)} \end{bmatrix}, \tag{A10}$$

where $O_{m\times n}$ denotes the null matrix of dimension $m \times n$.

The matrices of $[B]$ and $[\widehat{B}]$ can be defined as

$$[B]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times1} = \begin{bmatrix} [B]^{(1)}_{MN\times1} & [B]^{(2)}_{(2M+1)N\times1} & [B]^{(3)}_{M(S_1+1)\times1} & O_{M(S_2+1)\times1} & O_{MN\times1} & O_{MN\times1} \end{bmatrix}^T, \tag{A11 a}$$

and

$$[\widehat{B}]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times1} = \begin{bmatrix} O_{MN\times1} & O_{(2M+1)\times1} & O_{M(S_1+1)\times1} & [\widehat{B}]^{(4)}_{M(S_2+1)\times1} & O_{MN\times1} & O_{MN\times1} \end{bmatrix}^T, \tag{A11 b}$$

respectively.

$[\ ]_{j,J}$ represents the element of the jth row and Jth column of matrix.

he components of $[B]$ and $[\widehat{B}]$ are as follows

$$[B]^{(1)}_{j,1} = -\int_{-h}^{0}\int_{-l}^{l} \mathrm{e}^{\mathrm{i}k_y y}\overline{Y}_i(y)Z_1(z)Z_{j-N(i-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+N(i-1), Ni\right], \tag{A12 a}$$

$$[B]^{(2)}_{j,1} = \int_{-h}^{0}\int_{-l}^{l} \mathrm{i}k_x \mathrm{e}^{\mathrm{i}k_y y}Y^*_i(y)Z_1(z)Z_{j-N(i-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, 2M+1\right], j \in \left[1+N(i-1), Ni\right], \tag{A12 b}$$

$$[B]^{(3)}_{j,1} = \mathrm{i}kG\int_{-d_1}^{0}\int_{-l}^{l} \mathrm{e}^{\mathrm{i}k_y y}\overline{Y}_i(y)Z_1(z)u_{j-(S_1+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+(S_1+1)(i-1), (S_1+1)i\right], \tag{A12 c}$$

and

$$[\widehat{B}]^{(4)}_{j,1} = \left(-\frac{\mathrm{i}}{\rho\omega}\right)\Big/\left(-\frac{\mathrm{i}gA}{\omega}\right)\int_{-h}^{-d_2}\int_{-l}^{l} \overline{Y}_i(y)v_{j-(S_2+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+(S_2+1)(i-1), (S_2+1)i\right] \tag{A12 d}$$

matrices of $[X]$ and $[\widehat{X}]$, which represent the unknown coefficients, is given by

$$[X]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times 1} = \left[A^{(1)}_{(2M+1)N\times 1} \quad A^{(2)}_{MN\times 1} \quad A^{(3)}_{MN\times 1} \quad A^{(4)}_{MN\times 1} \quad H^{(1)}_{M(S_1+1)\times 1} \quad H^{(2)}_{M(S_2+1)\times 1}\right]^T, \tag{A13 a}$$

and

$$[\widehat{X}]_{[(2M+1)N+M(3N+S_1+S_2+2)]\times 1} = \left[\widehat{A}^{(1)}_{(2M+1)N\times 1} \quad \widehat{A}^{(2)}_{MN\times 1} \quad \widehat{A}^{(3)}_{MN\times 1} \quad \widehat{A}^{(4)}_{MN\times 1} \quad \widehat{H}^{(1)}_{M(S_1+1)\times 1} \quad \widehat{H}^{(2)}_{M(S_2+1)\times 1}\right]^T, \tag{A13 b}$$

respectively. We replace the unknowns $A^{(5)}_{mn}$ or $\widehat{A}^{(5)}_{mn}$ by the relation of $A^{(5)}_{mn} = A^{(4)}_{mn}\mathrm{e}^{2\overline{p}_{mn}a}$ or $\widehat{A}^{(5)}_{mn} = \widehat{A}^{(4)}_{mn}\mathrm{e}^{2\overline{p}_{mn}a}$ to reduce the dimension of the unknown matrices $[X]$ and $[\widehat{X}]$. The semi-analytical solution of the present wave-structure problem is rigidly verified using Haskind relation and energy conservation rule. It can be concluded that the omission of $A^{(5)}_{mn}$ or $\widehat{A}^{(5)}_{mn}$ will not affect the correctness of results.

The key factor affecting the computational efficiency of the present semi-analytical model is the size of the matrix (i.e., $[\Xi]$ in Eq. (A10) of Appendix A). The advantage of the present model in this paper, compared with the analytical model not handling the velocity singularity, is that the present model convergences at a relatively smaller truncated number. Correspondingly, the dimension of each sub-component in the total matrix $[\Xi]$ is determined by the truncated number. Although the auxiliary functions may increase the additional unknowns, the truncated number of the infinite series was reduced. In general, the introduction of the auxiliary functions of U_χ and V_χ (i.e., Eqs. (13) and (18)) is helpful to accelerate convergence.

The components of $[\Xi]$ are as follows

$$[\Xi]^{1,1}_{j,J} = \int_{-h}^{0}\int_{-l}^{l} Y_I(y)\overline{Y}_i(y)Z_{j-N(i-1)}(z)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+N(i-1), Ni\right], I \in \left[1, 2M+1\right], J \in \left[1+N(I-1), NI\right], \tag{A14 a}$$

$$[\Xi]^{1,2}_{j,J} = -\int_{-h}^{0}\int_{-l}^{l} \overline{Y}_I(y)\overline{Y}_i(y)Z_{j-N(i-1)}(z)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+N(i-1), Ni\right], I \in \left[1, M\right], J \in \left[1+N(I-1), NI\right], \tag{A14 b}$$

$$[\Xi]^{1,3}_{j,J} = -\int_{-h}^{0}\int_{-l}^{l} \overline{Y}_I(y)\overline{Y}_i(y)\mathrm{e}^{2b\overline{p}_{I,J-N(I-1)}}Z_{j-N(i-1)}(z)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+N(i-1), Ni\right], I \in \left[1, M\right], J \in \left[1+N(I-1), NI\right], \tag{A14 c}$$

$$[\Xi]^{1,5}_{j,J} = -\int_{-d_1}^{0}\int_{-l}^{l} \overline{Y}_I(y)\overline{Y}_i(y)u_{j-N(i-1)}(z)Z_{J-(S_1+1)(I-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+N(i-1), Ni\right], I \in \left[1, M\right], J \in \left[1+(S_1+1)(I-1), (S_1+1)I\right], \tag{A14 d}$$

$$[\Xi]^{2,1}_{j,J} = \int_{-h}^{0}\int_{-l}^{l} p_{I,J-N(I-1)}Y_I(y)Y^*_i(y)Z_{j-N(i-1)}(z)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, 2M+1\right], j \in \left[1+N(i-1), Ni\right], I \in \left[1, 2M+1\right], J \in \left[1+N(I-1), NI\right], \tag{A14 e}$$

$$[\Xi]^{2,2}_{j,J} = \int_{-h}^{0}\int_{-l}^{l} \overline{p}_{I,J-N(I-1)}\overline{Y}_I(y)Y^*_i(y)Z_{j-N(i-1)}(z)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, 2M+1\right], j \in \left[1+N(i-1), Ni\right], I \in \left[1, M\right], J \in \left[1+N(I-1), NI\right], \tag{A14 f}$$

$$[\Xi]^{2,3}_{j,J} = -\int_{-h}^{0}\int_{-l}^{l} \overline{p}_{I,J-N(I-1)}\mathrm{e}^{2b\overline{p}_{I,J-N(I-1)}}\overline{Y}_I(y)Y^*_i(y)Z_{j-N(i-1)}(z)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, 2M+1\right], j \in \left[1+N(i-1), Ni\right], I \in \left[1, M\right], J \in \left[1+N(I-1), NI\right], \tag{A14 g}$$

$$[\Xi]^{3,1}_{j,J} = -\mathrm{i}kG\int_{-d_1}^{0}\int_{-l}^{l} Y_I(y)\overline{Y}_i(y)Z_{J-N(I-1)}(z)u_{j-(S_1+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+(S_1+1)(i-1), (S_1+1)i\right], I \in \left[1, 2M+1\right], J \in \left[1+N(I-1), NI\right], \tag{A14 h}$$

$$[\Xi]^{3,2}_{j,J} = \int_{-d_1}^{0}\int_{-l}^{l} \left(-\overline{p}_{I,J-N(I-1)} + \mathrm{i}kG\right)\overline{Y}_I(y)\overline{Y}_i(y)Z_{J-N(I-1)}(z)u_{j-(S_1+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i \in \left[1, M\right], j \in \left[1+(S_1+1)(i-1), (S_1+1)i\right], I \in \left[1, M\right], J \in \left[1+N(I-1), NI\right], \tag{A14 i}$$

$$[\Xi]^{3,3}_{j,J} = \int_{-d_1}^{0}\int_{-l}^{l}\left(\overline{p}_{I,J-N(I-1)}e^{2b\overline{p}_{I,J-N(I-1)}} + \mathrm{i}kG\right)\overline{Y}_I\left(y\right)\overline{Y}_i(y)Z_{J-N(I-1)}\left(z\right)u_{j-(S_1+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+(S_1+1)(i-1),(S_1+1)i\right], I\in\left[1,M\right], J \in\left[1+N(I-1),NI\right], \tag{A14 j}$$

$$[\Xi]^{4,2}_{j,J} = \int_{-h}^{-d_2}\int_{-l}^{l}e^{2b\overline{p}_{I,J-N(I-1)}}\overline{Y}_I\left(y\right)\overline{Y}_i(y)v_{J-N(I-1)}(z)Z_{j-(S_2+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+(S_2+1)(i-1),(S_2+1)i\right], I\in\left[1,M\right], J\in\left[1+N(I-1),NI\right], \tag{A14 k}$$

$$[\Xi]^{4,3}_{j,J} = \int_{-h}^{-d_2}\int_{-l}^{l}\overline{Y}_I\left(y\right)\overline{Y}_i(y)v_{J-N(I-1)}(z)Z_{j-(S_2+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+(S_2+1)(i-1),(S_2+1)i\right], I\in\left[1,M\right], J\in\left[1+N(I-1),NI\right], \tag{A14 l}$$

$$[\Xi]^{4,4}_{j,J} = -\int_{-h}^{-d_2}\int_{-l}^{l}\left[1+e^{4a\overline{p}_{I,J-N(I-1)}}\right]\overline{Y}_I\left(y\right)\overline{Y}_i(y)v_{J-N(I-1)}(z)Z_{j-(S_2+1)(i-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+(S_2+1)(i-1),(S_2+1)i\right], I\in\left[1,M\right], J \in\left[1+N(I-1),NI\right], \tag{A14 m}$$

$$[\Xi]^{5,2}_{j,J} = \int_{-h}^{0}\int_{-l}^{l}\left(-\overline{p}_{I,J-N(I-1)}\right)\left[e^{2b\overline{p}_{I,J-N(I-1)}}\right]\overline{Y}_I\left(y\right)\overline{Y}_i(y)Z_{j-N(i-1)}\left(z\right)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+N(i-1),Ni\right], I\in\left[1,M\right], J\in\left[1+N(I-1),NI\right], \tag{A14 n}$$

$$[\Xi]^{5,3}_{j,J} = \int_{-h}^{0}\int_{-l}^{l}\overline{p}_{I,J-N(I-1)}\overline{Y}_I\left(y\right)\overline{Y}_i(y)Z_{j-N(i-1)}\left(z\right)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+N(i-1),Ni\right], I\in\left[1,M\right], J\in\left[1+N(I-1),NI\right], \tag{A14 o}$$

$$[\Xi]^{5,6}_{j,J} = -\int_{-h}^{-d_2}\int_{-l}^{l}\overline{Y}_I\left(y\right)\overline{Y}_i(y)v_{j-N(i-1)}\left(z\right)Z_{J-(S_2+1)(I-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+N(i-1),Ni\right], I\in\left[1,M\right], J\in\left[1+(S_2+1)(I-1),(S_2+1)I\right], \tag{A14 p}$$

$$[\Xi]^{6,4}_{j,J} = \int_{-h}^{0}\int_{-l}^{l}\left(-\overline{p}_{I,J-N(I-1)}\right)\left[1-e^{4a\overline{p}_{I,J-N(I-1)}}\right]\overline{Y}_I\left(y\right)\overline{Y}_i(y)Z_{j-N(i-1)}\left(z\right)Z_{J-N(I-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+N(i-1),Ni\right], I\in\left[1,M\right], J \in\left[1+N(I-1),NI\right], \tag{A14 q}$$

and

$$[\Xi]^{6,6}_{j,J} = -\int_{-h}^{-d_2}\int_{-l}^{l}\overline{Y}_I\left(y\right)\overline{Y}_i(y)v_{j-N(i-1)}\left(z\right)Z_{J-(S_2+1)(I-1)}(z)\mathrm{d}y\mathrm{d}z, i\in\left[1,M\right], j\in\left[1+N(i-1),Ni\right], I\in\left[1,M\right], J\in\left[1+(S_2+1)(I-1),(S_2+1)I\right]. \tag{A14 r}$$

Appendix B

The computational efficiency is determined by the matrix size of $[\Xi]$ (see Eq. (A 10)), whose row number and column number are denoted by R and C, respectively. $R = (2M + 1)N + M(3N + S_1+S_2+2)$ and $C = (2M + 1)N + M(3N + S_1+S_2+2)$ for the present model. Section 2.4.1 shows that the present solution with Galerkin technique converges at truncated number $N = 40$ (correspondingly, $R = 1100$ and $C = 1100$). The truncated number of convergence test for the model with and without Galerkin technique is shown in Table B1. It can be found that the conventional method without Galerkin technique (i.e., the method similar to Zhang et al. (2021)) calls for a larger truncated number ($N \geq 110$) to achieve convergence. This is because the velocity singularity (order $-1/2$) at the tip of the chamber wall leads to poor convergence. The thickness of the chamber wall, the truncated number of expansion series in y-direction, truncated number of auxiliary function at the perforated wall are set as $0.0005h$, 5, and 5, respectively, for the method without the Galerkin method. Fig. B1 shows the matrix size, characterized by $R \times C$, corresponding to the semi-analytical method with and without the Galerkin technique. It is evident that, under the premise that the solution is converged, the size of $[\Xi]_{R\times C}$ for model without Galerkin method (i.e., 3900×3900, $N = 110$) is much larger than using Galerkin method (i.e., 1100×1100, $N = 40$). Hence, the present semi-analytical solution is more efficient by the introduction of Galerkin technique to accelerate model convergence.

Table B1
Reflection coefficient K_r calculated with and without Galerkin technique for truncated number N from 30 to 120. The geometrical and wave parameters are fixed as $\theta = \pi/4$, $b/h = 0.25$, $a/h = 0.25$, $d_1/h = 0.4$, $d_2/h = 0.6$, $G = 0.5(1 + 1\mathrm{i})$, and $l/h = 0.5$. For cases with Galerkin technique, the other truncated number are fixed as $M = 5$, $S_1 = 5$, and $S_2 = 5$.

Truncated number N	Reflection coefficient K_r						
	$kh = 1$	$kh = 1.5$	$kh = 2$	$kh = 2.5$	$kh = 3$	$kh = 3.5$	$kh = 4$
30	0.290	0.333	0.480	0.490	0.560	0.404	0.525
50	0.282	0.352	0.487	0.493	0.561	0.403	0.526
70	0.278	0.359	0.489	0.494	0.562	0.402	0.527
90	0.276	0.363	0.489	0.494	0.562	0.401	0.527

(continued on next page)

Table B1 (*continued*)

Truncated number N	Reflection coefficient K_r						
	$kh = 1$	$kh = 1.5$	$kh = 2$	$kh = 2.5$	$kh = 3$	$kh = 3.5$	$kh = 4$
110	0.274	0.365	0.489	0.494	0.564	0.400	0.527
120	0.274	0.365	0.489	0.494	0.564	0.400	0.527
Present model ($N = 40$)	0.274	0.365	0.489	0.494	0.564	0.400	0.527

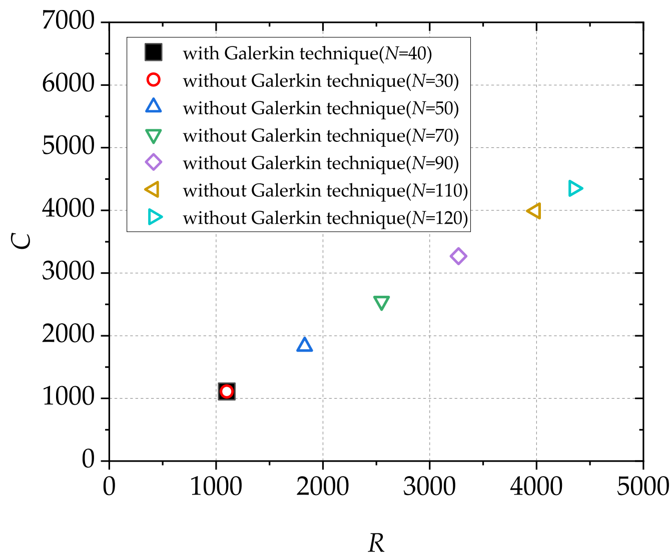

Fig. B1. The size of $[\Xi]_{R\times C}$ for the semi-analytical model with and without Galerkin technique.

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